



Biodiversity risk and firms' access to trade credit

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ABSTRACT

This study examines the relationship between firm-level exposure to biodiversity risk and access to supplier financing. We find that firms' access to trade credit decreases significantly with increased firm-level exposure to biodiversity risk. Mechanism test shows that reduced operating performance is the primary channel through which biodiversity risk affects access to trade credit. This effect is more pronounced among firms with low market power, those in industries with higher exposure to biodiversity risk, and those with weak corporate culture or low product innovation. Moreover, we find that firms highly exposed to biodiversity risk that receive less trade credit from suppliers tend to extend less trade credit to their customers. Given the growing attention to biodiversity risk, our findings offer important implications for both policymakers and corporate decision-makers seeking to understand and manage the broader financial and operating consequences of biodiversity risk.

1. Introduction

Biodiversity refers to the wide range of living organisms within the ecosystems and their associated ecological complexes. The degradation of biodiversity resources poses a significant threat to the global economy, that is distinct from climate-related risk (Adamolekun, 2024; Bach et al., 2024; de Carvalho et al., 2023; Giglio et al., 2023).² Unlike climate change risk, which reflects the negative exposure to adverse climate events (Fava et al., 2024; Sautner et al., 2023; Wang et al., 2022), biodiversity risk relates to firms' exposure to the potential downsides associated with damage to ecosystems and other related ecological losses (Adamolekun, 2024; Kedward et al., 2023).³ According to the World Economic Forum's Global Risks Report (WEF, 2024), biodiversity loss is ranked as the third-largest risk threatening society over the next decade. A recent report from the World Wide Fund for Nature indicates that the average population of more than 5,000 vertebrate species has declined by approximately 73 % between 1970 and 2020 (WWF, 2024). Similarly, a recent report from the World Bank provides alarming statistics on the economic losses linked to biodiversity and

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E-mail addresses: ksalmaghrabi@kau.edu.sa (K.S. Almaghrabi), walid.benamar@udst.edu.qa (W. Ben-Amar), kongziyu@cuhk.edu.cn (Z. Kong).¹ Walid Ben-Amar gratefully acknowledges the financial support of the Telfer Research Fellowship.² Biodiversity risk even surpasses climate change risk as a significant challenge to sustainable finance (Karolyi & Tobin-de la Puente, 2023). See Adamolekun (2024), Bach et al. (2024), de Carvalho et al. (2023), and Giglio et al. (2023) for a discussion of the differences between the two forms of risk.³ For example, firms dependent on specific natural resources, such as pharmaceutical firms, may face disruption in their R&D process or drug production due to loss of biodiversity resources or habitat hazards. Similarly, firms operating within fishery industry may encounter severe shortages as a result of ecosystem losses (Kedward et al., 2023).

ecosystem degradation (World Bank, 2021). Statistics also indicate that if current trends continue, between 30 % and 50 % of all species are expected to be lost by the middle of this century (Deutz et al., 2020). As firms depend on biodiversity resources and contribute to environmental degradation, they are exposed to biodiversity risk (Giglio et al., 2023; Houdet et al., 2012). This risk comprises both physical risks (i.e., the direct economic losses caused by biodiversity degradation such as disruption to supply chains and production processes) as well as transition risks (i.e., the uncertainty surrounding regulation changes, policy shifts, legal fines, or the potential for litigation). Both result in loss of firms' economic activity and assets' value (Giglio et al., 2023).

Despite the significant impact of biodiversity risk on firms' operations and the controversies it raises (Dempsey, 2013; Panwar, 2023), few studies have explored this issue. Moreover, only a limited number of studies have investigated its influence at the firm level. Therefore, we respond to recent calls by Cosma et al. (2023), Karolyi and Tobin-de la Puente (2023), and Hutchinson and Lucey (2024) for research that explores the different effects of biodiversity by examining whether firms' biodiversity risk exposure influences their access to trade credit.

Trade credit represents a significant portion of corporate financing, surpassing the volume of other forms of short-term financing (Hasan & Habib, 2019; Petersen & Rajan, 1997). Its use has also remarkably increased over the past 50 years (Li et al., 2023). For example, Yang and Birge (2018) highlight that trade credit accounts for approximately 30 % more than bank loans in terms of usage, while Levine et al. (2018) find that, on average, trade credit represents about 25 % of a firm's total debt liabilities. Trade credit enables firms to overcome liquidity constraints and improve technical efficiency (Agostino & Trivieri, 2019; D'Mello & Toscano, 2020). As a form of supplier-provided financing, it is offered in the context of product market transactions, with customer firms committing to pay at a later date. Unlike financial institutions and other stakeholders, suppliers often possess superior operational insights into their customer firms' business practices, supply chains, production processes, and risk exposures (Burkart & Ellingsen, 2004; Fabbri & Menichini, 2010). This informational advantage makes trade credit decisions particularly sensitive to risks that directly affect firm operations, including those arising from biodiversity degradation.

When extending trade credit, suppliers are particularly concerned with their customer firms' ability to fulfill their obligations (Klapper et al., 2012; Zhang et al., 2020) and the potential for customer firms to transmit financial shocks to them if they are unable to repay the trade credit (Hertzel et al., 2008; Jacobson & von Schedvin, 2015; Lian, 2017; Raddatz, 2010). Consequently, suppliers' willingness to extend credit and the duration of the credit terms offered are contingent upon these considerations (Klapper et al., 2012; Zhang et al., 2020). Due to its effect on firms' operating performance and the disruptions it could cause to supply chains and production processes, biodiversity risk has the potential to influence these considerations, making trade credit a leading indicator of how biodiversity risk is being factored into inter-firm financial relations. Furthermore, as an informal and decentralized credit channel, trade credit plays a crucial role in shaping how supply chain finance transmits and amplifies firm-level shocks.

We argue that firms exposed to biodiversity risk are less likely to receive trade credit, as suppliers perceive such exposure as a signal of a reduced capacity to meet financial obligations and an increased risk of financial contagion. However, the complexity of biodiversity risk may pose a challenge for suppliers in assessing the effects of exposure to such risk. Moreover, there is evidence that biodiversity risk is not equally priced by all stakeholders. As suppliers often possess better information regarding a firm's credit-worthiness (Burkart & Ellingsen, 2004; Fabbri & Menichini, 2010), they could be more willing to extend credit when firms are exposed to high biodiversity risk. Given these competing perspectives, it is important to empirically examine the impact of such risk on access to trade credit.

Utilizing the text-based measure of firm-level biodiversity risk provided by Giglio et al. (2023), we find that firms' access to trade credit from suppliers (measured as accounts payable divided by the cost of goods sold and, alternatively, by purchases) decreases with increased firm-level exposure to biodiversity risk. This effect is both statistically and economically significant. Moreover, the results hold when addressing the endogeneity issue using an instrumental variable (IV) estimation, difference-in-differences (DiD) regression and analysis based on supplier-customer matched pairs. In addition, our results are robust when (1) ruling out the alternative explanation of decreased trade credit demand; (2) using alternative biodiversity risk measures; (3) using alternative trade credit measures; and (4) including additional control variables.

Our mechanism test shows that reduced operating performance is the primary channel through which biodiversity risk affects access to trade credit. In addition, we conduct several cross-sectional tests to shed further light on the effect of biodiversity risk exposure on trade credit. First, we posit that a buyer's market power may allow them to negotiate longer payment periods (Fabbri & Klapper, 2016; Gonçalves et al., 2018). Accordingly, firms exposed to biodiversity risk with greater market power may be able to obtain more trade credit from suppliers. Using different measures of product market competition, we find that suppliers provide more trade credit to firms exposed to biodiversity risk when they have greater market power. We also examine whether trade credit reduction is more pronounced for firms in industries with higher biodiversity risk exposure. We find that biodiversity risk exposure is associated with less trade credit from suppliers only when firms are in highly exposed industries, suggesting that suppliers consider industry norms when making trade credit decisions.

Furthermore, we investigate factors that may alleviate the adverse effects of biodiversity risk. We conjecture that more transparent information environments and long-term orientation may mitigate stakeholder-agency conflicts, thereby enabling firms to secure more trade credit. We examine whether a strong corporate culture, which reflects greater corporate transparency and a long-term orientation, affects the relationship between biodiversity risk exposure and trade credit. We find that the relationship between biodiversity risk exposure and the receipt of trade credit is less pronounced for firms with stronger corporate culture. Moreover, we examine the role of product innovation in mitigating the relationship between biodiversity risk exposure and access to trade credit. Suppliers may also consider firms' growth opportunities and the role of product innovation in addressing transition biodiversity risks when determining trade credit terms for highly exposed firms. Our evidence highlights that product innovation alleviates the negative effect of biodiversity risk exposure on access to trade credit.

Lastly, we investigate whether the reduction in trade credit financing spills over throughout the supply chain. Specifically, we examine whether firms with high biodiversity risk exposure, who receive less trade credit from their suppliers, are also less likely to extend generous trade credit terms to their own customers. Our results confirm this prediction, suggesting that biodiversity risk exposure results in a trade credit spillover effect within the supply chain.

We contribute to two main streams of literature. First, we contribute to the emerging studies on biodiversity risk, which document that such risk affects stock returns (Garel et al., 2024; Giglio et al., 2023; Ma et al., 2024), firm efficiency (Li et al., 2025), future stock price crashes (Liang et al., 2024), bond yield spreads (Soylemezgil & Uzmanoglu, 2024), firm-level profitability (Créti et al., 2024), bankruptcy risk (Adamolekun, 2024), and overall firm performance (Bach et al., 2024). We contribute to these studies by showing that biodiversity risk exposure also affects firms' access to trade credit. This finding is important given the role of trade credit as a primary source of short-term financing (Hasan & Habib, 2019; Petersen & Rajan, 1997).

Second, we contribute to studies on the determinants of trade credit, which highlight multiple influencing factors, including firm size (Lawrenz & Oberndorfer, 2018), CEO reputation (Quan et al., 2024), business group affiliation (Chong & Im, 2024), earnings quality (García-Teruel et al., 2014), bargaining power (Fabbri & Klapper, 2016; Gonçalves et al., 2018), financial reporting quality (X. Li et al., 2021), the level of regional trust (Levine et al., 2018; Wu et al., 2014), and national culture (El Ghoul & Zheng, 2016), among others. In particular, we build on the growing body of research focusing on the effects of climate-related risks on trade credit (Cao et al., 2024; Liu et al., 2025; Zaman et al., 2025). For example, Cao et al. (2024) demonstrate that suppliers reduce the provision of trade credit in response to heightened climate uncertainty. Liu et al. (2025) show that exposure to climate-related physical risks, either from suppliers or customers, reduces customer firms' access to trade credit. Similarly, Zaman et al. (2025) find that firms with strong green credentials, such as those adopting renewable energy, enjoy better access to trade credit. We extend the findings of these studies by identifying biodiversity risk as a distinct and important determinant of firms' access to supplier financing. We document that firms' access to trade credit significantly decreases with heightened firm-level exposure to biodiversity risk, even after accounting for climate-related risks. Hence, our findings highlight the importance of distinguishing biodiversity risk from climate risk in studies of trade credit and inter-firm financial relationship.

Given the growing importance of biodiversity risk over the past decade (Dempsey, 2013), our findings are both timely and relevant, offering several practical implications for policymakers and corporate decision-makers. First, this study provides important insights into recent global initiatives to address the biodiversity crisis, such as the Kunming Declaration (2021), the Kunming-Montreal Global Biodiversity Framework (GBF, 2022), and the Task Force on Nature-related Financial Disclosures (TNFD, 2023). For example, the GBF calls for urgent and transformative action by governments at all levels to meet global biodiversity goals by 2030.⁴⁵ It introduces institutional and policy targets that encourage the integration of biodiversity considerations into business operations and inter-firm financial practices. These efforts aim to create regulatory certainty and new market opportunities for companies adopting nature-friendly strategies.

As governments begin to align national regulations and investment flows with the GBF targets (Targets 14, 18, 19), suppliers are likely to face increased pressure to assess and mitigate their exposure to firms that may undermine biodiversity compliance along the value chain. This is particularly emphasized by the GBF's Target 5, which explicitly calls for enhanced transparency regarding firms' biodiversity exposure, its impacts on operations, supply and value chains, and portfolios, as well as measures undertaken to reduce such exposure. Similarly, the TNFD provides an overview of biodiversity indices to measure businesses' impacts on biodiversity. Our results indicate that suppliers consider biodiversity-related risks when extending trade credit, offering new insights into the heightened awareness of biodiversity risk not only among investors but also among other stakeholders. This underscores a growing recognition of both physical and transition risks in corporate outcomes. In light of emerging regulations and frameworks like the GBF and TNFD, our results suggest that suppliers may further restrict trade credit to customers with poor biodiversity performance, particularly when such exposure could result in reputational damage or regulatory penalties.

Second, our results demonstrate that firm-level exposure to biodiversity risk affects access to trade credit, with the effect being more pronounced in industries that are highly dependent on nature and ecosystem services. Therefore, this study has significant implications for policymakers, as it emphasizes the importance of considering the impact of biodiversity risk both at the firm and industry levels when designing regulatory frameworks related to this issue. Rather than adopting broad, one-size-fits-all policies, regulators may consider implementing sector-specific guidelines that reflect the unique dependencies and risk profiles of different industries. Such an approach could improve compliance feasibility and reduce disproportionate burdens on sectors more acutely exposed to biodiversity-related risks, ultimately supporting more effective biodiversity risk mitigation strategies across the corporate landscape.

Lastly, the findings reveal a complex interplay between environmental considerations and the stability of the supply chain. On the one hand, suppliers appear to incorporate firms' biodiversity risk as a factor in their decisions regarding the provision of trade credit. On the other hand, the reduction in trade credit financing spills over throughout the supply chain as exposed firms that receive less supplier financing also extend less trade credit to their own customers. This implies an effect on both the upstream and downstream of the supply chain, and, in turn, represents a threat to the stability of the supply chain. This finding suggests that firms need to consider how to manage biodiversity risk and foster collaborative relationships with suppliers. This is especially important for firms with low market power and for those in industries with high biodiversity risk exposure, as these risks further impede their ability to access trade credit. Given prior studies indicating that effective management of biodiversity risk can mitigate its negative consequences (Bassen

⁴ The framework was adopted by over 185 countries at the United Nations Biodiversity Conference (COP15) on December 19, 2022.

⁵ The framework also includes four overarching long-term objectives to be achieved by 2050.

et al., 2024), along with recent calls and initiatives to address the biodiversity crisis, our findings highlight that it is increasingly essential for firms to integrate biodiversity into their corporate strategies and actions. While certain sectors may unavoidably impact species and ecosystems, our results emphasize the need to adopt “biodiversity offsets”, which involve conservation activities designed to compensate for lasting impacts in order to achieve an “no net loss” outcome (Giglio et al., 2024).

The remainder of the paper is structured as follows. Section 2 reviews previous literature on biodiversity risk and sets the developed hypotheses, while section 3 indicates the source of data used in our analysis and discusses our research design. Section 4 presents our results on the effect of biodiversity risk exposure on trade credit, addresses the endogeneity issue, and conducts a series of sensitivity tests, as well as mechanism, cross-sectional, and additional analyses. Section 5 concludes the study.

2. Literature review and hypothesis development

The degradation of biodiversity poses a significant threat to the global economy. According to the World Wide Fund for Nature, global populations of animals, fish, birds, reptiles, and amphibians have declined by approximately 69 % over the last few decades (WWF, 2022). Moreover, a recent World Bank report indicates that the continued loss of biodiversity and the associated decline in ecosystem services could result in an annual loss of \$2.7 trillion in global GDP over the next five years (World Bank, 2021). It also suggests that more than half of the global GDP is either partially or fully dependent on nature and ecosystem services. The necessity of conserving natural capital is further underscored by the difficulty of offsetting or compensating for environmental damage, given the irreplaceable value of these resources (Giglio et al., 2023; Guer et al., 2024).

As an integral part of society, firms can be directly affected by biodiversity-related risks, as the loss of biodiversity adversely affects economic activities. Biodiversity physical risk (i.e., the direct economic losses caused by biodiversity degradation) can disrupt supply chains and production processes, leading to shortages of raw materials and operational disruption within firms (Li et al., 2025; Ma et al., 2024). For example, firms dependent on specific natural resources, such as pharmaceutical firms, may face disruption in their research and development (R&D) process or drug production due to loss of biodiversity resources or habitat hazards.⁶ This risk is particularly critical as firms are increasingly reliant on natural resources. For instance, in the French context, Svartzman et al. (2021) find that approximately 42 % of the value of securities held is owned by issuers that are significantly dependent on ecosystem services.

Additionally, the uncertainty surrounding future biodiversity-related transition risks, stemming from regulation changes, policy shifts, legal fines, or the potential for litigation, poses a significant challenge for firms, including changes in asset values and reputational damage (Garel et al., 2024; Giglio et al., 2023; Liang et al., 2024; Ma et al., 2024). For example, firms with activities that could harm ecosystems (such as energy firms with drilling and exploration activities or utility firms that produce solar or wind energy) face biodiversity risks arising from potential related regulations, which could result in substantial economic losses for these firms (Giglio et al., 2023).⁷ These costs are expected to be high, given the increased attention towards biodiversity preservation (Garel et al., 2024).

Furthermore, biodiversity conservation efforts require substantial financial resources (Flammer et al., 2024), making firms' exposure to biodiversity risk costly. de Carvalho et al. (2023) note that around 30 % of firms that have implemented biodiversity policies over the past four years have an enterprise value of approximately \$7.2 trillion exposed to unmanaged biodiversity risk. This situation may necessitate firms exposed to greater biodiversity risk to hold additional capital reserves to sustain their operations in the event that biodiversity risk materializes (Li et al., 2025).

Considering the potential effects of biodiversity, recent and ongoing studies have begun to investigate the impact of such risk on stock returns and other aspects of firms' operational and financial performance. Karolyi and Tobin-de la Puente (2023) propose a framework that highlights the risks associated with biodiversity loss, explores how these risks may be priced, and offers suggestions on how to raise the investment needed to address biodiversity loss. Giglio et al. (2023) examine whether firm-level biodiversity exposure influences economic activity and asset prices, finding that biodiversity risk adversely affects stock prices. Kalhor and Kyaw (2024) document that firms effectively managing biodiversity-related risks experience positive abnormal returns in the days leading up to the 2021 Kunming Declaration and the 2022 UN Biodiversity Conference. Similarly, using an international sample, Garel et al. (2024) find that biodiversity footprints do not explain cross-sectional differences in stock returns prior to the release of the Kunming Declaration. Nevertheless, they find that investors require a higher risk premium for firms with greater biodiversity footprints in the years preceding the event. In the Chinese context, Ma et al. (2024) find that firm-level biodiversity risk is negatively correlated with stock returns. Based on the potential impact of biodiversity risk on the volatility of future profitability and cash flows, Liang et al. (2024) suggest that biodiversity risk is linked to stock crash risk, providing consistent evidence that firm-level exposure to biodiversity risk increases the likelihood of future stock price crashes. Adamolekun (2024) documents that biodiversity risk is linked to an increased likelihood of bankruptcy, and that its impact is distinct from the effects of climate change risk on bankruptcy likelihood. Soylemezgil and Uzmanoglu (2024) document some evidence that firms with higher biodiversity risk incur higher bond yield spreads. Li et al. (2025) find that biodiversity risk exposure negatively impacts firms' efficiency in terms of external financing and cost of capital. Cr  ti et al. (2024) find that firm-level profitability is lower in countries experiencing a trend of declining biodiversity. Bach et al. (2024) document that firm-level biodiversity exposure negatively impacts overall firm performance.

Overall, evidence from recent and ongoing research indicates that investors deem biodiversity exposure to be a unique risk factor, necessitating compensation in the form of a risk premium due to the potential physical risk and uncertainty surrounding future biodiversity policies (i.e., transition risk). Furthermore, findings from recent studies suggest that exposure to biodiversity loss

⁶ Newman and Cragg (2020) note that approximately 60% of all new chemicals are produced based on or extracted from natural resources.

⁷ Dasgupta (2021) provides a review of the economic effects of biodiversity.

adversely affects firms' operations and overall performance. However, there is limited evidence on the effect of biodiversity risk exposure on firms' access to trade credit.

According to the Stakeholder theory (Freeman, 1984; Freeman & Reed, 1983), firms interact closely with a diverse array of stakeholders, including customer, suppliers and regulators, who significantly influence a firm's ability to achieve its objectives (Freeman, 1984). Therefore, firms must adhere to specific principles and ensure that their actions align with normative commitments to maintain a good relationship with their stakeholders (Donaldson & Preston, 1995). Furthermore, the theory posits that since firms are composed of these stakeholders, their actions impact stakeholders' responses and decisions (Clarkson, 1994, 1995). When considering biodiversity risk, different stakeholders may pay varying levels of attention and hold diverse perceptions of this risk (Adamolekun, 2024) based on their interactions with firms and the types of claims they hold.⁸ Suppliers, as critical stakeholders who provide trade credit to firms, are particularly sensitive to environmental and operational risks that could impact their customers' ability to repay or maintain stable business relationships. In this context, empirical research indicates that suppliers often function as informal monitors of firm behavior and risk exposure (Fabbri & Klapper, 2016; Giannetti et al., 2011).

Specifically, suppliers offer trade credit based on product market transactions, in which customer firms promise to pay within a specified period. This promise constitutes an implicit financing contract, in which the suppliers take the risk that the customer firms will not pay in the future, and that they face the future default risk of customers. Consequently, when extending trade credit, suppliers are concerned about customer firms' abilities to fulfill their obligations (Klapper et al., 2012; Zhang et al., 2020). As such, signals of firms' future profitability influence the trade credit granted by suppliers. For example, Zhang et al. (2020) find that corporate social responsibility, which reflects a greater ability to meet financial obligations, is positively associated with access to trade credit. Quan et al. (2024) document that firms with better CEO reputation, proxied by the receipt of prestigious awards, tend to obtain significantly more trade credit from suppliers. Moreover, when extending trade credit, suppliers are concerned about the potential of customer firms to transmit financial shocks to them if they are unable to repay their trade credit (Hertzel et al., 2008; Jacobson & von Schedvin, 2015; Lian, 2017; Raddatz, 2010). Accordingly, suppliers' willingness to provide credit and the duration of the credit period they extend depend on these factors (Klapper et al., 2012; Zhang et al., 2020).

Building on prior studies showing that biodiversity risk exposure negatively affects firms' financial and operating performance, we conjecture that such exposure would also affect the stability of trade credit terms across the supply chain. Specifically, we expect upstream firms to be less willing to extend trade credit to downstream firms with high biodiversity risk exposure, as the latter could have less capability to fulfill their financial obligations. Additionally, downstream firms exposed to biodiversity risk could have a higher probability of transmitting negative biodiversity (physical and transitional) shocks through the supply chain to suppliers (upstream firms). The influence of biodiversity risk exposure on access to raw materials and disruption to operations might severely affect firm production and sales, hindering their ability to repay trade credit.

For instance, the recent *Biodiversity Insights Report*, by *Textile Exchange* (2021), compiles survey findings and company-level insights from the textile and fashion sector. It highlights suppliers' recognition of biodiversity risk as a central issue that affects the health of their raw-material supply base. Numerous participating companies noted that habitat degradation and declining ecosystem health could cause production disruptions, impair long-term resource access, and increase financial risks across the value chain. These findings reinforce the notion that biodiversity risk is not merely an environmental issue but a material operational concern for supply chain partners. Hence, considering these factors along with stakeholder theory, a firm's biodiversity risk, which represents deviation from normative commitments to ecosystem services, may lead suppliers to respond by reducing trade credit to avoid financial or reputational spillovers. Based on the potential negative effect of biodiversity risk on access to trade credit, we formulate our first hypothesis as follows:

Hypothesis 1a. Biodiversity risk is negatively associated with access to trade credit.

However, biodiversity risk may have a minimal effect on trade credit. Despite the potential adverse impacts of biodiversity on the stock market and firm performance, a survey conducted by Giglio et al. (2023) highlights that biodiversity risk is not adequately priced in financial markets. Similarly, Arnold and Tran (2024) document that biodiversity exposure does not influence bond ratings assigned by rating agencies. Nedopil (2023) argues that market participants may not fully account for the costs associated with biodiversity risk in their financial decision-making. Cr  ti et al. (2024) provide evidence of the underpricing of biodiversity-related physical risks by market participants due to inadequate analysis of the impact of such risks on a firm's future cash flow.

Moreover, this underpricing may paradoxically lead to increased trade credit access. Mispricing and unexpected cash flow shocks may prompt suppliers to extend more trade credits particularly if they are motivated by a desire to preserve supply chain stability (Ersahin et al., 2024). Additionally, some firms might make limited efforts to mitigate their biodiversity impacts, which might result in higher earnings and returns (Garel et al., 2024). This, in turn, might allow them to have better access to trade credit from their suppliers. Based on this potential positive effect of biodiversity risk on access to trade credit, we formulate the following competing hypothesis:

Hypothesis 1b. Biodiversity risk is positively associated with access to trade credit.

⁸ For example, prior studies document divergent (and at times conflicting) responses to biodiversity risk from different stakeholders. While some studies indicate a negative impact of biodiversity risk on shareholders (e.g., Giglio et al., 2023; Liang et al., 2024) and bondholders (Soylemezgil & Uzmanoglu, 2024), other papers document limited effect of such risk on credit rating agencies (Arnold & Tran, 2024). Furthermore, studies by Cr  ti et al. (2024) and Nedopil (2023) show that biodiversity risk is often underpriced by market participants.

3. Research design

3.1. Data and sample construction

We obtain our data from multiple datasets. All firm-level financial information data for trade credit and control variables are from Compustat Annual North American file. Data on firm-level exposure to biodiversity risk are retrieved from Giglio et al. (2023).⁹ We construct an instrumental variable (IV) using data from Google Trends for our Two-Step Least Squares (2SLS) analysis.¹⁰ For cross-sectional analyses, we obtain the product market competition data from Hoberg and Phillips (2016),¹¹ industry-level biodiversity risk exposures from Giglio et al. (2023), corporate culture data from K. Li et al. (2021),¹² product life cycle data from Hoberg and Maksimovic (2022),¹³ and supplier-customer relation data from Compustat Segment Customer File.

Our sample selection begins with listed U.S. firms that have available data on firm-level biodiversity measures from fiscal years 2000 to 2023. We then merge these firm-year observations with the required Compustat data for calculating trade credit and control variables. After excluding firms in financial (4-digit SIC code: 6000–6999) and utilities (4-digit SIC code: 4900–4999) industries as they are highly regulated, our final sample consists of 34,431 firm-year observations corresponding to 4,005 unique firms.

3.2. Variable Definition

3.2.1. Access to trade credit

Following the extant literature (Hasan & Habib, 2019; Levine et al., 2018; X. Li et al., 2021; Zhang et al., 2020), we measure access to trade credit using accounts payable to the cost of goods sold (COGS) (*Payable/COGS*) and accounts payable to purchases (*Payable/Purchases*). Purchases are calculated as COGS plus year-to-year changes in inventory.

3.2.2. Firm-Level biodiversity risk exposure

We measure biodiversity risk exposure using firm-level data from Giglio et al. (2023), who employ textual analysis on 10-K filings to construct firm-specific biodiversity risk measures. These measures include: (1) *Count*, an indicator coded as one if a company mentions biodiversity in at least two sentences in its 10-K filing; (2) *Negative*, the difference between the number of negative biodiversity sentences and positive biodiversity sentences; and (3) *Regulation*, an indicator equaling one if a company mentions biodiversity in at least two sentences, with at least one referring to regulation. Giglio et al. (2023) use a dictionary-based method to identify biodiversity- and regulation-related terms in 10-Ks. They employ a deep learning language model (BERT) (Devlin et al., 2019) to classify sentences containing biodiversity terms into positive, neutral, or negative sentiments.

We use *Biodiversity Negative* as our primary variable of interest. It represents the net number of sentences in a firm's disclosures that highlight negative biodiversity risks. This measure is specifically designed to distinguish biodiversity risks from opportunities, thereby providing a more accurate reflection of the nature and extent of firm-level exposure to biodiversity risk. In robustness checks, we also use two additional variables: *Biodiversity Count*, which captures the overall frequency of biodiversity-related mentions, and *Biodiversity Regulation*, which specifically identifies references to regulatory risks associated with biodiversity in 10-K filings. In addition, we also adopt nature dependence scores developed by Garel et al. (2025) as an alternative proxy of a firm's exposure to biodiversity risk and use it in our robustness checks to address the potential bias of disclosure-based measurement.

3.2.3. Control variables

We incorporate a vector of control variables that have been identified in prior literature as being related to access to trade credit (e.g., Abdulla et al., 2020; Cao et al., 2024; Chong & Im, 2024; D'Mello & Toscano, 2020; García-Teruel et al., 2014; Quan et al., 2024). Corporate financing patterns differ significantly between smaller and larger firms (Beck et al., 2008). Larger buyers are more likely to receive more trade credit from smaller suppliers due to their higher bargaining power (Klapper et al., 2012). Thus, we control for firm size (*Ln(Assets)*), measured as the natural logarithm of total assets. Additionally, we include a control for firm age using the natural logarithm of firm age (*Ln(Age)*). Older firms generally enjoy stronger financial conditions and easier access to external funding (Abdulla et al., 2020; Gertler, 1988), reducing their reliance on alternative financing channels. To account for the potential trade-off between investing in fixed assets and working capital (Abdulla et al., 2020; Quan et al., 2024), we control for fixed assets measured as the net property, plant, and equipment to total assets (*PPE/Assets*). Suppliers are likely to consider not only the net profit margin on current sales but also the present value of the net profit margin on future sales when deciding whether to offer trade credit to customers (Petersen & Rajan, 1997). To account for this, we control for sales growth (*ΔSales*), measured as the percentage change in sales over year *t* to year *t-1*. We also account for firm profitability using return on assets (*ROA*) as more profitable firms are typically perceived as more reliable and financially stable by suppliers, making them more likely to receive favorable trade credit terms. We further account for firms' investment opportunities by including the market-to-book ratio (*MB*) as higher growth opportunities might increase firm risk

⁹ Data are available at <https://www.biodiversityrisk.org/download/>.

¹⁰ We adopt a python package *pytrends* to obtain the measures of Google search series for the terms “biodiversity loss”, “ecosystem services”, and “species loss”, following Giglio et al. (2023).

¹¹ Data are available at <https://hobergphillips.tuck.dartmouth.edu/industryconcen.htm>.

¹² Data are available at <https://sites.google.com/view/kaili/finance-publications>.

¹³ Data are available at <https://faculty.marshall.usc.edu/Gerard-Hoberg/HobergMaxLifeCycles/index.html>.

(Fama & French, 1993). Suppliers may factor in these risks when determining whether to extend trade credit. Finally, to account for a firm's internal financial conditions, which suppliers may consider when making credit-granting decisions, we include leverage (*Debts/Assets*) measured as the total debts to total assets, cash holdings (*Cash/Assets*) measured as cash and short-term investments to total assets, and cash flow volatility ($\sigma\text{Cash Flow}$) measured as the standard deviation of cash flow over at least 3 of the last 5 years (Quan et al., 2024). Appendix A specifies the details of variable definitions.

3.3. Regression model

To test our hypothesis, we regress the measures of trade credit on biodiversity risk as well as a battery of controls. The baseline Ordinary Least Square (OLS) regression model is formalized as follows:

$$AP_{i,t} = \alpha + \beta_b \text{Biodiversity}_{i,t} + \beta_x \mathbf{X}_{i,t} + \delta_i + \nu_t + \varepsilon_{i,t}, \quad (1)$$

where $AP_{i,t}$ is the measures of trade credit from suppliers for firm i at year t (*Payable/COGS* and *Payable/Purchases*); $\text{Biodiversity}_{i,t}$ is the measure of biodiversity risk exposure (*Biodiversity Negative*) of firm i at year t . $\mathbf{X}_{i,t}$ denotes a battery of controls as defined in Section 3.2.3 and Appendix A. We include firm and year fixed effects, and cluster standard errors at the firm level.

4. Empirical results

4.1. Descriptive statistics

Table 1 reports the descriptive statistics, with all continuous variables winsorized at the 1st and 99th percentiles. Panel A shows that the mean (median) reported level of accounts payable to COGS in our sample is 0.157 (0.109), which is similar to the results of Abdulla et al. (2020). Notably, the standard deviation indicates relatively high volatility, primarily driven by the upper quartile, which highlights significant cross-sectional differences in firms' access to trade credit. The mean (median) value of *Biodiversity Negative* is 0.018 (0.000). An average firm in our sample has approximately \$6,255 million in total assets, an age of 62 years, 24.2 % fixed assets, 15.7 % sales growth, a slightly positive return on assets (ROA), a market-to-book ratio of 3.763, 24.8 % leverage, 21.2 % cash holdings (cash-to-assets), and cash flow volatility of 0.071.

We report sample distribution by year and industry (Fama-French 48 (FF48) industry classifications) in Panels B and C. The number of firm-year observations shows a roughly monotonic increase from 2013, peaking in 2021. Notably, the number of observations decreases after 2000, as well as after 2007, likely due to the dotcom bust and the global financial crisis, respectively. Business Services is the most represented industry in our sample, while industries such as Fabricated Products, Textiles, and Tobacco Products are among the least represented.¹⁴

4.2. Baseline results

Table 2 reports the results of estimating Equation (1). Columns (1) and (2) present estimations that do not include control variables. The two columns show that biodiversity risk exposure is significantly associated with lower access to trade credit from suppliers, at the 1 % significance level. Columns (3) and (4) present consistent results after incorporating the control variables. Specifically, we find that the coefficient of biodiversity risk exposure is still negative and significant at the 1 % level. This finding suggests that firms with higher exposure to biodiversity risk are less likely to receive trade credit from suppliers. Economically, a one-point increase in biodiversity risk exposure results in a 0.036 (0.025) decrease in accounts payable relative to COGS (Purchases), representing 17.4 % (12.1 %) of its standard deviation or \$142.9 million (\$100.3 million) for an average firm in dollar terms.^{15,16} These findings support Hypothesis 1a.

Control variables load as expected. For example, sales growth is positively associated with accounts payable to COGS (purchases), consistent with prior studies (e.g., D'Mello & Toscano, 2020). This finding suggests that suppliers consider both current and future profit margins when deciding whether to extend trade credit (Petersen & Rajan, 1997). Additionally, higher leverage is associated with reduced trade credit from suppliers, indicating that firms with greater debt are less likely to receive trade credit from suppliers. Lastly, cash holdings are negatively correlated with trade credit, suggesting that firms with higher cash reserves, often maintained for precautionary motives, may face reduced access to trade credit, potentially due to the positive association between cash reserves and long-term default risk (Acharya et al., 2012).

¹⁴ Our sample distribution is consistent with prior studies on trade credit (e.g., Jory et al., 2020) and closely mirrors the year and industry distributions observed in the broader Compustat universe over the corresponding period.

¹⁵ On average, firms in our sample report a COGS of \$3,969 million and purchases of \$4,011 million.

¹⁶ We do not interpret the economic magnitude based on a one-standard-deviation increase in negative biodiversity risk exposure, as the measure is integer-based (i.e., a one-point increase indicates a one-sentence increase in net negative disclosures). However, such an interpretation would still yield meaningful results (4.1% (3.1%) based on the standard deviation or \$33.7 million (\$23.72 million) for an average firm in dollar terms).

Table 1
Descriptive statistics.

Panel A: Descriptive Statistics of Full Sample								
	Observations	Mean	Std.Dev	Min	P25	Median	P75	Max
Payable/COGS	34,431	0.157	0.208	0.000	0.065	0.109	0.166	1.535
Payable/Purchases	34,059	0.151	0.191	0.000	0.064	0.107	0.162	1.385
Biodiversity Negative	34,431	0.018	0.236	−5.000	0.000	0.000	0.000	6.000
Ln(Assets)	34,431	7.187	1.728	3.499	5.922	7.084	8.319	11.683
Ln(Age)	34,431	4.120	0.114	3.912	4.025	4.127	4.220	4.290
PPE/Assets	34,431	0.242	0.222	0.004	0.074	0.166	0.345	0.885
ΔSales	34,431	0.157	0.415	−0.601	−0.001	0.080	0.208	2.781
ROA	34,431	0.000	0.212	−1.090	−0.010	0.047	0.094	0.370
MB	34,431	3.763	7.021	−24.717	1.513	2.523	4.393	43.969
Debts/Assets	34,431	0.248	0.213	0.000	0.059	0.223	0.371	0.976
Cash/Assets	34,431	0.212	0.236	0.001	0.040	0.117	0.298	0.950
σCash Flow	34,431	0.071	0.126	0.004	0.015	0.030	0.067	0.861
Panel B: Sample Distribution over Years								
Year	Frequency				Percentage (%)			
2000	1418				4.12			
2001	1259				3.66			
2002	1213				3.52			
2003	1295				3.76			
2004	1407				4.09			
2005	1392				4.04			
2006	1440				4.18			
2007	1479				4.30			
2008	1380				4.01			
2009	1296				3.76			
2010	1356				3.94			
2011	1343				3.90			
2012	1323				3.84			
2013	1418				4.12			
2014	1474				4.28			
2015	1481				4.30			
2016	1429				4.15			
2017	1427				4.14			
2018	1470				4.27			
2019	1458				4.23			
2020	1581				4.59			
2021	1834				5.33			
2022	1693				4.92			
2023	1565				4.55			
Total:	34,431				100			
Panel C: Sample Distribution across Industry (FF48)								
Fama-French 48 industries	Frequency				Percentage (%)			
Fabricated Products	66				0.19			
Textiles	85				0.25			
Tobacco Products	87				0.25			
Agriculture	89				0.26			
Coal	89				0.26			
Precious Metals	90				0.26			
Candy & Soda	114				0.33			
Defense	114				0.33			
Beer & Liquor	126				0.37			
Shipbuilding, Railroad Equipment	136				0.39			
Shipping Containers	180				0.52			
Recreation	203				0.59			
Non-Metallic and Industrial Metal Mining	209				0.61			
Printing and Publishing	235				0.68			
Rubber and Plastic Products	271				0.79			
Aircraft	293				0.85			
Almost Nothing	292				0.85			
Business Supplies	401				1.16			
Entertainment	476				1.38			
Personal Services	491				1.43			
Electrical Equipment	509				1.48			
Apparel	524				1.52			
Steel Works Etc.	531				1.54			

(continued on next page)

Table 1 (continued)

Consumer Goods	671	1.95
Construction Materials	712	2.07
Construction	714	2.07
Automobiles and Trucks	727	2.11
Healthcare	733	2.13
Food Products	760	2.21
Restaurants, Hotels, Motels	817	2.37
Measuring and Control Equipment	830	2.41
Communication	1018	2.96
Computers	1057	3.07
Chemicals	1066	3.10
Transportation	1123	3.26
Wholesale	1376	4.00
Medical Equipment	1393	4.05
Machinery	1505	4.37
Petroleum and Natural Gas	1614	4.69
Retail	2076	6.03
Electronic Equipment	2270	6.59
Pharmaceutical Products	3407	9.90
Business Services	4951	14.38
Total:	34,431	100

This table reports the descriptive statistics. Panel A reports the descriptive statistics of the full sample; Panel B reports the sample distribution over years; Panel C reports the sample distribution across industry (Fama-French 48 Industries) sorting by the ascending order of frequencies. All continuous variables are winsorized at 1st and 99th percentiles. *Payable/COGS* is accounts payable to cost of goods sold (COGS); *Payable/Purchases* is accounts payable to purchases, where purchases are measured as COGS plus the change of inventory; *Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences; $\ln(\text{Assets})$ is the natural logarithm of total assets; $\ln(\text{Age})$ is the natural logarithm of firm age, measured as the years since a firm appears in Annual Compustat North American file; *PPE/Assets* is the net property, plant, and equipment to total assets; ΔSales is the percentage change in sales over year t to year $t-1$; *ROA* is return on total assets; *MB* is market value of equity to book value of equity; *Debts/Assets* is total debts to total assets; *Cash/Assets* is cash and short-term investments to total assets; $\sigma\text{Cash Flow}$ is standard deviation of cash flows over at least 3 of the last 5 years. See Appendix A for variable definitions.

4.3. Identification strategies

4.3.1. Instrument variable Method: Two-Stage least Squares (2SLS) regression

Our baseline results can be subject to the endogeneity problem due to unobservable firm characteristics. Firms' exposure to biodiversity risk and the access to trade credit can be simultaneously determined by unobservable firm characteristics such as indifferent corporate culture towards biodiversity or sustainability issues. Such a culture may lead to inadequate management of toxic byproducts, causing significant harm to local biodiversity. On the other hand, it can worsen stakeholder relationships, potentially affecting firm access to trade credit. Therefore, following [Bach et al. \(2024\)](#), we first instrument for biodiversity risk exposure using the Google Biodiversity Attention Index ([Giglio et al., 2023](#)). Specifically, we aggregate attention indexes that analyze the annual Google searches for the terms "biodiversity loss", "ecosystem services", and "species loss". Google users' attention to biodiversity risks reflects the overall severity of these risks and the level of public awareness. This public attention is linked to firm-level exposure to biodiversity risk; however, it should not directly influence trade credit, except through firm-specific risk channel. In other words, firms with insignificant biodiversity risk exposure are unlikely to be affected by public attention to these issues.

[Table 3](#) shows the 2SLS regression results. As shown in Columns (1) and (2), the aggregated Google Attention Index is negatively associated with biodiversity risk in the first-stage analysis of the 2SLS regressions. The negative coefficients contrast with the results of [Bach et al. \(2024\)](#), which is not surprising given that their analysis uses a regulation-related measure of biodiversity risk, differing from our measure of negative biodiversity risk exposure. Specifically, heightened public attention to biodiversity risk may positively influence the development of relevant legislation and regulatory risk, while reducing negative risk exposure, as increased public awareness and stricter regulations help mitigate local biodiversity loss. Columns (3) and (4) show that the coefficients of the instrumented biodiversity risk measure remain significantly negative, indicating that our baseline results are robust to the endogeneity problem from unobservable firm characteristics. Notably, the Cragg-Donald Wald F -statistics (51.164 and 34.900) exceed the rule-of-thumb threshold ($F = 10$) suggested by [Staiger and Stock \(1997\)](#), as well as the critical values of [Stock and Yogo \(2005\)](#), indicating that our instrument qualifies as "strong" in statistical terms.

4.3.2. Difference-in-Difference analysis

Second, to further address the endogeneity issue, we exploit the 2019 regulatory rollback of the Endangered Species Act (ESA) as an exogenous shock and employ a difference-in-differences (DiD) approach as a complementary identification strategy. The ESA is a U.S.

Table 2
Results of baseline regressions.

	(1) Payable/COGS	(2) Payable/Purchases	(3) Payable/COGS	(4) Payable/Purchases
Biodiversity Negative	−0.037*** (−2.759)	−0.026*** (−2.726)	−0.036*** (−2.665)	−0.025*** (−2.706)
Ln(Assets)			0.005 (0.996)	0.005 (1.041)
Ln(Age)			0.005 (0.180)	0.019 (0.823)
PPE/Assets			−0.015 (−0.472)	−0.001 (−0.046)
ΔSales			0.027*** (4.799)	0.015*** (3.175)
ROA			−0.008 (−0.448)	−0.009 (−0.602)
MB			0.000 (1.190)	0.000 (0.706)
Debts/Assets			−0.025 (−1.333)	−0.030* (−1.779)
Cash/Assets			−0.041** (−2.229)	−0.018 (−1.086)
σCash Flow			−0.038 (−1.090)	−0.028 (−0.848)
Intercept	0.164*** (33.832)	0.154*** (34.454)	0.125 (1.439)	0.057 (0.716)
Firm	✓	✓	✓	✓
Year	✓	✓	✓	✓
Observations	34,431	34,059	34,431	34,059
Adj. R ²	0.645	0.662	0.648	0.663

This table reports the results of estimating Equation (1) with Ordinary Least Squares (OLS) regressions. *Payable/COGS* is accounts payable to cost of goods sold (COGS); *Payable/Purchases* is accounts payable to purchases, where purchases are measured as COGS plus the change of inventory; *Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences; *Ln(Assets)* is the natural logarithm of total assets; *Ln(Age)* is the natural logarithm of firm age, measured as the years since a firm appears in Annual Compustat North American file; *PPE/Assets* is the net property, plant, and equipment to total assets; *ΔSales* is the percentage change in sales over year t to year $t-1$; *ROA* is return on total assets; *MB* is market value of equity to book value of equity; *Debts/Assets* is total debts to total assets; *Cash/Assets* is cash and short-term investments to total assets; *σCash Flow* is standard deviation of cash flows over at least 3 of the last 5 years. See Appendix A for variable definitions. t -statistics are in parentheses. *** 1 % significance; ** 5 % significance; * 10 % significance.

federal law designed to protect endangered and threatened species, as well as their habitats. Its primary aim is to prevent extinction and to promote the recovery of at-risk species. On August 12, 2019, the Trump administration substantially weakened the ESA by introducing changes to its implementation.¹⁷ These changes significantly weakened protections for threatened species and allowed federal agencies to conduct economic analyses when deciding whether to extend protections to a specie (Lambert, 2019). In September 2019, a coalition of 18 state attorneys general filed a lawsuit to block the ESA rollbacks, pledging to maintain existing protections to endangered species.^{18,19} We posit that these 18 states are more inclined to resist the rollback and maintain the *status quo* of species protection, while others are more likely to be affected by the policy changes and face biodiversity downgrades (i.e., higher biodiversity risk). Therefore, we estimate DiD regressions based on the ESA rollback in a $[-4, 4]$ window.

$$AP_{i,t} = \alpha + \beta_t Treat_i + \beta_p Post_t + \beta_{tp} Treat_i \times Post_t + \beta_x X_{i,t} + \delta_i + \nu_t + \varepsilon_{i,t}, \quad (2)$$

where $AP_{i,t}$ is the measures of trade credit from suppliers for firm i at year t (*Payable/COGS* and *Payable/Purchases*); $Treat_i$ is an indicator coding one if a firm locates in non-suing states; $Post_t$ is an indicator coding one for the years 2019–2023, zero for the years 2015–2018. $X_{i,t}$ denotes a battery of controls as defined in Section 3.2.3 and Appendix A. We include firm and year fixed effects, and cluster standard errors at the firm level.

Table 4 shows that the estimated coefficient $\hat{\beta}_p$ is significantly negative across the two models, suggesting that firms headquartered in non-suing states are more likely to experience constrained access to trade credit after the ESA rollback in 2019. Fig. 1 further shows no evidence of a violation of the parallel trend assumption.

¹⁷ The changes introduced by the Trump administration have been partially cancelled by the Biden administration in March 2024. See <https://www.nytimes.com/2024/03/28/climate/endangered-species-rules-regulations.html>.

¹⁸ See <https://stateimpactcenter.org/news-events/press-releases/esa-lawsuit>.

¹⁹ These states are Colorado, Connecticut, Illinois, Michigan, Nevada, New Jersey, New Mexico, New York, North Carolina, Oregon, Pennsylvania, Rhode Island, Vermont, Washington, the District of Columbia, Maryland, Massachusetts, and California.

Table 3
Results of 2SLS regressions.

	First Stage		Second Stage	
	(1) Biodiversity Negative	(2) Biodiversity Negative	(3) Payable/COGS	(4) Payable/Purchases
Google Attention	−0.041*** (−3.394)	−0.032*** (−3.172)		
BiodiversityNegative			−0.424** (−2.203)	−0.431** (−2.000)
Ln(Assets)	0.009* (1.739)	0.008 (1.645)	0.009 (1.590)	0.009* (1.665)
Ln(Age)	0.240*** (2.932)	0.187** (2.577)	0.037 (1.265)	0.046* (1.692)
PPE/Assets	0.085*** (2.950)	0.085*** (2.883)	0.026 (0.759)	0.036 (1.119)
ΔSales	−0.005 (−1.312)	−0.004 (−1.052)	0.024*** (4.079)	0.014*** (2.811)
ROA	−0.037*** (−2.636)	−0.029** (−2.368)	0.004 (0.183)	0.004 (0.222)
MB	−0.000** (−2.152)	−0.000** (−2.030)	0.000 (1.291)	0.000 (0.494)
Debts/Assets	−0.000 (−0.012)	−0.006 (−0.389)	−0.024 (−1.115)	−0.034* (−1.757)
Cash/Assets	0.019 (1.361)	0.020 (1.386)	−0.037* (−1.926)	−0.014 (−0.793)
σCash Flow	0.084** (2.335)	0.069* (1.934)	−0.022 (−0.579)	−0.021 (−0.602)
Intercept	−1.020*** (−3.099)	−0.803*** (−2.794)	−0.049 (−0.464)	−0.094 (−0.952)
Firm	✓	✓	✓	✓
Observations	29,246	28,898	29,246	28,898
F-Statistics	51.164	34.900		
Adj. R ²	0.005	0.004		

This table reports the results of estimating two-stage least squares (2SLS) regressions with instrumental variables (*Google Attention*). *Payable/COGS* is accounts payable to cost of goods sold (COGS); *Payable/Purchases* is accounts payable to purchases, where purchases are measured as COGS plus the change of inventory; *Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences; *Google Attention* is Google Biodiversity Attention Index (Giglio et al., 2023), measured as the aggregated attention indexes that analyses the annual Google searches for the terms “biodiversity loss”, “ecosystem services”, and “species loss”, scaled by 1000; *Ln(Assets)* is the natural logarithm of total assets; *Ln(Age)* is the natural logarithm of firm age, measured as the years since a firm appears in Annual Compustat North American file; *PPE/Assets* is the net property, plant, and equipment to total assets; *ΔSales* is the percentage change in sales over year *t* to year *t-1*; *ROA* is return on total assets; *MB* is market value of equity to book value of equity; *Debts/Assets* is total debts to total assets; *Cash/Assets* is cash and short-term investments to total assets; *σCash Flow* is standard deviation of cash flows over at least 3 of the last 5 years. See Appendix A for variable definitions. *t*-statistics are in parentheses. *F*-statistics is the Cragg-Donald Wald *F*-statistics. *** 1 % significance; ** 5 % significance; * 10 % significance.

4.3.3. Supplier-Customer pairs

Third, to further validate our findings, we conduct a more granular analysis at the supplier–customer level, leveraging disclosure requirements under SEC Regulation S-K, in which firms must disclose any customer accounting for more than 10 % of total sales. We follow prior studies (e.g., Barrot & Sauvagnat, 2016; Campello & Gao, 2017; Houston et al., 2016) and collect data from the Compustat Segment Customer File to construct a sample of matched customer–supplier pairs.

For the consistency with our main hypothesis that focal firms’ biodiversity risk affects their access to trade credit, we examine how a focal firm’s (firm *i*) biodiversity risk influences the trade credit decisions of its suppliers (firm *j*). More specifically, we match each firm in our sample to its suppliers and collect data to calculate the suppliers’ characteristics. In addition, we require available sales data for each supplier–customer pair to calculate a customer’s share of its supplier’s total sales (Ding et al., 2024), and biodiversity risk data for suppliers to control for the effect of suppliers’ own biodiversity risk. Our final sample comprises 1,034 suppliers and 390 customers, resulting in 7,917 supplier–customer–year observations.

Following prior studies (e.g., Ding et al., 2024), we estimate the following regression model:

$$\text{SupplierAR}_{j,t} = \alpha + \beta_{cb} \text{CustomerBiodiversity}_{i,t} + \beta_{sc} \text{SupplierCharacteristics}_{j,t} + \beta_{cc} \text{CustomerCharacteristics}_{i,t} + \delta_{ij} + \nu_{i,t} + \theta_{j,t} + \varepsilon_{ij,t}, \quad (3)$$

where $\text{SupplierAR}_{j,t}$ is the measure of trade credit provision for supplier *j* at year *t* (*Supplier Receivable/Sales*); $\text{CustomerBiodiversity}_{i,t}$ is the measure of biodiversity risk exposure (*Customer Biodiversity Negative*) for customer *i* at year *t*; $\text{SupplierCharacteristics}_{j,t}$ and $\text{CustomerCharacteristics}_{i,t}$ denotes vectors of supplier and customer characteristics as defined in Section 3.2.3 and Appendix A. We further incorporate suppliers’ own biodiversity risk and customers’ share of sales (Ding et al., 2024). We include pair fixed effects to control for unobservable, time-invariant factors specific to each supplier–customer relationship, as well as supplier (customer) industry-by-year fixed effects, which absorb the confounding effects of time-varying shocks within a supplier’s and a customer’s FF48

Table 4
Results of difference-in-differences analysis.

	(1) Payable/COGS	(2) Payable/Purchases
Post	0.072** (2.324)	0.025 (0.953)
Treat × Post	−0.012** (−2.143)	−0.009* (−1.731)
Ln(Assets)	0.002 (0.288)	0.002 (0.373)
Ln(Age)	−0.426 (−1.538)	−0.016 (−0.068)
PPE/Assets	0.009 (0.325)	0.007 (0.287)
ΔSales	0.024*** (2.884)	0.014** (2.133)
ROA	−0.006 (−0.260)	−0.009 (−0.491)
MB	−0.000 (−0.168)	−0.000 (−0.961)
Debts/Assets	0.016 (0.633)	−0.005 (−0.204)
Cash/Assets	−0.075*** (−2.908)	−0.054** (−2.397)
σCash Flow	−0.028 (−0.569)	−0.057 (−1.325)
Intercept	1.911* (1.662)	0.199 (0.200)
Firm	✓	✓
Year	✓	✓
Observations	13,938	13,736
Adj. R ²	0.685	0.709

This table reports the results of estimating difference-in-differences analysis using the rollback of the Endangered Species Act (ESA) by the Trump administration in 2019 as an exogenous shock. *Payable/COGS* is accounts payable to cost of goods sold (COGS); *Payable/Purchases* is accounts payable to purchases, where purchases are measured as COGS plus the change of inventory; *Treat* is an indicator coding one if a firm locates in non-suing states; *Post* is an indicator coding one for the years 2019–2023, zero for the years 2015–2018; *Ln(Assets)* is the natural logarithm of total assets; *Ln(Age)* is the natural logarithm of firm age, measured as the years since a firm appears in Annual Compustat North American file; *PPE/Assets* is the net property, plant, and equipment to total assets; *ΔSales* is the percentage change in sales over year *t* to year *t*-1; *ROA* is return on total assets; *MB* is market value of equity to book value of equity; *Debts/Assets* is total debts to total assets; *Cash/Assets* is cash and short-term investments to total assets; *σCash Flow* is standard deviation of cash flows over at least 3 of the last 5 years. See Appendix A for variable definitions. *t*-statistics are in parentheses. *** 1 % significance; ** 5 % significance; * 10 % significance.

industry, as well as industry-level differences between suppliers and customers. Standard errors are clustered at supplier level.

Table 5 shows that the coefficients of customers' biodiversity risk are significantly negative across different specifications. This evidence from supplier-customer matched pairs further supports the conclusion that suppliers take *their* customers' biodiversity risk into account when determining trade credit provision.

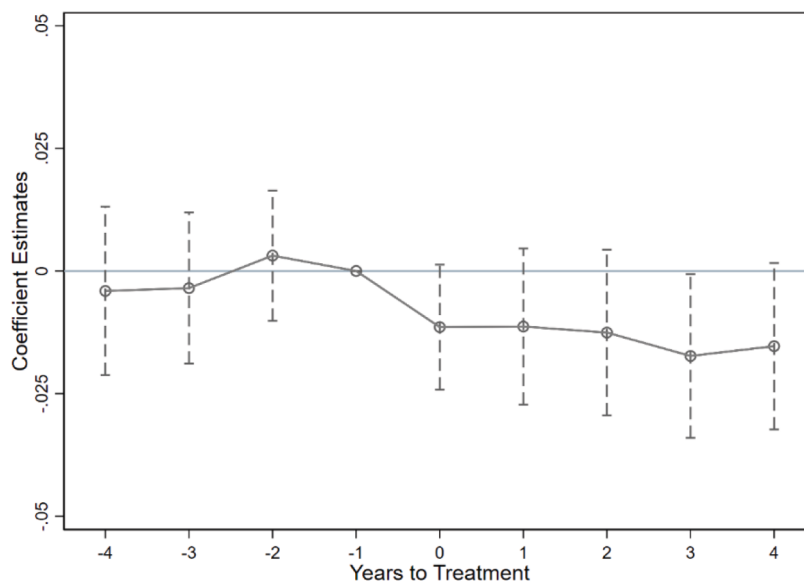
4.4. Additional robustness checks

In this subsection, we conduct additional tests to confirm the robustness of our results. Specifically, we consider whether our results remain robust when we (1) rule out the alternative explanation that our baseline results are driven by the reduced demand for trade credit; (2) use alternative biodiversity risk measures; (3) use adjusted trade credit measures; and (4) include additional control variables and fixed effects.

First, the reduced access to trade credit among highly exposed firms may also stem from these firms increasing their cash holdings as a precautionary response to potential negative impacts, thereby reducing their demand for trade credit.²⁰ This alternative explanation confounds the mechanism whereby suppliers take firms' biodiversity risk into account when making trade credit decisions. To address this potential explanation of our results, we partition the sample based on the median of corporate cash holdings (*Cash/Assets*) and rerun our baseline regressions for the subsamples. If the alternative explanation holds, we would expect to observe a more pronounced effect of biodiversity risk on access to trade credit among firms with higher cash holdings. Panel A of Table 6 shows that the

²⁰ We thank an anonymous reviewer for suggesting the consideration of demand-side effect.

Dependent Variable: Payable/COGS



Dependent Variable: Payable/Purchases

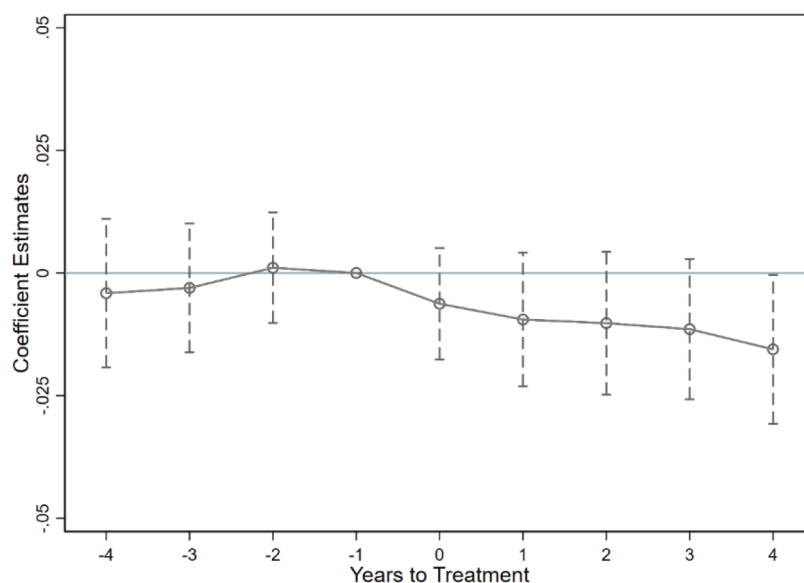


Fig. 1. Parallel Trends. This figure plots the coefficients of event-study difference-in-differences regressions and their 95 % confidence intervals for firms headquartered in non-suing states with indicators capturing the relative time to and from the Endangered Species Act (ESA) by the Trump administration in 2019. Year -1 has been omitted as the reference year.

coefficient of biodiversity risk is significant and negative only when cash holdings are low, and the Wald test shows that the difference between the coefficients for the high- and low-cash groups is not significant. This disproves the possibility that firms with higher biodiversity risk facing reduced access to trade credit when they hold more cash. Therefore, this finding helps to mitigate the concern that the observed effect may be driven by high-risk firms decreasing their demand for trade credit due to accumulating higher cash holdings.²¹

²¹ Untabulated analyses yield qualitatively similar results when we partition the sample based on excess cash holdings, calculated according to the specification of Opler et al. (1999). In addition, we find no evidence of an association between biodiversity risk and subsequent increases in cash holdings.

Table 5
Results of analysis at the supplier-customer pair level.

	(1) Supplier Receivable/Sales	(2) Supplier Receivable/Sales	(3) Supplier Receivable/Sales
Customer Biodiversity Negative	−0.028** (−2.430)	−0.024** (−2.181)	−0.025** (−2.240)
Supplier Ln(Assets)		0.010** (2.046)	0.011** (2.219)
Supplier PPE/Assets		−0.216*** (−6.767)	−0.214*** (−6.516)
Supplier ΔSales		−0.014** (−1.973)	−0.014** (−1.979)
Supplier ROA		−0.055*** (−2.660)	−0.055*** (−2.660)
Supplier MB		0.004* (1.837)	0.004* (1.881)
Supplier Debts/Assets		−0.009 (−0.564)	−0.008 (−0.486)
Supplier Cash/Assets		−0.133*** (−6.196)	−0.134*** (−6.208)
Supplier σCash Flow		0.032 (0.617)	0.036 (0.687)
Supplier Biodiversity Negative		0.002 (0.393)	0.002 (0.381)
Customer Ln(Assets)			−0.012 (−1.631)
Customer PPE/Assets			0.011 (0.345)
Customer ΔSales			0.009 (1.143)
Customer ROA			−0.052* (−1.744)
Customer MB			0.003 (0.972)
Customer Debts/Assets			−0.001 (−0.060)
Customer Cash/Assets			−0.016 (−0.493)
Customer σCash Flow			0.048 (0.411)
Customer Share			0.021 (0.784)
Intercept	0.162*** (1806.352)	0.162*** (4.444)	0.270*** (3.473)
Supplier-Customer Pair	✓	✓	✓
Supplier Industry × Year	✓	✓	✓
Customer Industry × Year	✓	✓	✓
Observations	7917	7917	7917
Adj. R ²	0.675	0.697	0.698

This table reports the results of analysis at the supplier-customer pair level. *Supplier Receivable/Sales* is suppliers' accounts receivable to sales; *Customer (Supplier) Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences for customers (suppliers). *Customer (Supplier) Ln(Assets)* is the natural logarithm of total assets for customers (suppliers); *Customer (Supplier) PPE/Assets* is the net property, plant, and equipment to total assets for customers (suppliers); *Customer (Supplier) ΔSales* is the percentage change in sales over year t to year $t-1$ for customers (suppliers); *Customer (Supplier) ROA* is return on total assets for customers (suppliers); *Customer (Supplier) MB* is market value of equity to book value of equity for customers (suppliers); *Customer (Supplier) Debts/Assets* is total debts to total assets for customers (suppliers); *Customer (Supplier) Cash/Assets* is cash and short-term investments to total assets for customers (suppliers); *Customer (Supplier) σCash Flow* is standard deviation of cash flows over at least 3 of the last 5 years for customers (suppliers). *Customer Share* is a supplier's sales to a specific customer to the supplier's total sales. See Appendix A for variable definitions. t -statistics are in parentheses. *** 1 % significance; ** 5 % significance; * 10 % significance.

Columns (1) to (4) of Panel B show that our results still hold when using alternative biodiversity risk measures, including the general mention of biodiversity and regulatory biodiversity risk developed by Giglio et al. (2023). This suggests that the effect of biodiversity risk exposure on the trade credit received from suppliers is unlikely to be driven exclusively by the measure of negative sentiment. In addition, our results may be subject to bias from the disclosure-based nature of the biodiversity risk measure developed by Giglio et al. (2023). Specifically, under a voluntary disclosure regime, managers may choose to selectively or belatedly disclose information related to biodiversity risk. To address the potential measurement issues, we employ the natural dependence measure developed by (Garel et al., 2025) as an alternative proxy of a firm's exposure to biodiversity risk. This score captures firm-level dependency on nature and ecosystem services—those where disruptions could result in functionality loss and/or financial adaptation

Table 6

Other robustness checks.

Panel A: Subsample Analysis Based on Cash Holdings						
	High Cash/Assets	Low Cash/Assets	High Cash/Assets	Low Cash/Assets		
	(1) Payable/COGS	(2) Payable/COGS	(3) Payable/Purchase	(4) Payable/Purchase		
Biodiversity Negative	−0.004 (−0.124)	−0.039** (−2.544)	−0.002 (−0.076)	−0.027*** (−2.640)		
Intercept	0.307* (1.745)	0.029 (0.287)	0.210 (1.338)	−0.038 (−0.408)		
Baseline Controls	✓	✓	✓	✓		
Firm	✓	✓	✓	✓		
Year	✓	✓	✓	✓		
Wald Test (p-value)	0.225		0.286			
Observations	17,215	17,216	17,081	16,978		
Adj. R ²	0.576	0.761	0.598	0.764		
Panel B: Alternative Biodiversity Measures						
	(1) Payable/COGS	(2) Payable/Purchases	(3) Payable/COGS	(4) Payable/Purchases	(5) Payable/COGS	(6) Payable/Purchases
Biodiversity Mention	−0.037** (−2.430)	−0.028** (−2.325)				
Biodiversity Regulation			−0.041** (−1.961)	−0.034* (−1.876)		
Nature Dependence					−0.022* (−1.760)	−0.020* (−1.658)
Intercept	0.103 (1.195)	0.040 (0.506)	0.108 (1.237)	0.043 (0.529)	−0.412*** (−3.934)	−0.339*** (−3.413)
Baseline Controls	✓	✓	✓	✓	✓	✓
Firm	✓	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓	✓
Observations	34,134	33,770	34,134	33,770	8663	8556
Adj. R ²	0.647	0.662	0.647	0.662	0.845	0.850
Panel C: Alternative Trade Credit Measures						
	(1) Adj. Payable/ COGS	(2) Adj. Payable/ Purchases	(3) Payable/COGS (t + 1)	(4) Payable/Purchases (t + 1)	(5) Residual Payable/ COGS	(6) Residual Payable/ Purchases
Biodiversity Negative	−0.036*** (−2.665)	−0.025*** (−2.706)	−0.046*** (−2.622)	−0.030** (−2.510)	−0.026* (−1.946)	−0.017** (−2.008)
Intercept	0.017 (0.193)	−0.050 (−0.621)	−0.071 (−0.794)	−0.067 (−0.789)	−0.474** (−2.301)	−0.506** (−2.577)
Baseline Controls	✓	✓	✓	✓	✓	✓
Firm	✓	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓	✓
Observations	34,431	34,059	29,398	29,083	27,615	27,282
Adj. R ²	0.637	0.652	0.663	0.680	0.626	0.644
Panel D: Additional Controls						
	(1) Payable/COGS			(2) Payable/Purchases		
Biodiversity Negative	−0.031** (−2.017)			−0.018** (−1.998)		
CC Exposure	−3.671 (−0.256)			−7.909 (−0.619)		
FA Maturity	0.004*** (4.291)			0.003*** (3.721)		
Turnover	−0.000 (−0.905)			−0.000 (−0.688)		
Capx/Assets	0.124* (1.959)			0.100* (1.713)		
R&D/Assets	0.016 (0.168)			0.052 (0.687)		
Financing Costs	−0.012* (−1.652)			−0.008 (−1.261)		
Life Cycle	−0.001 (−1.577)			−0.001 (−1.176)		

(continued on next page)

Table 6 (continued)

Panel D: Additional Controls		
	(1) Payable/COGS	(2) Payable/Purchases
Financial Constraints	0.000* (1.793)	0.000* (1.868)
Bankruptcy Risk	−0.001 (−0.921)	−0.001 (−1.139)
Intercept	−0.031** (−2.017)	−0.018** (−1.998)
Baseline Controls	✓	✓
Firm	✓	✓
Year	✓	✓
Observations	24,326	24,026
Adj. R ²	0.683	0.700
Panel E: Alternative Fixed Effects		
	(1) Payable/COGS	(2) Payable/Purchases
Biodiversity Negative	−0.036*** (−2.885)	−0.026*** (−3.048)
Intercept	0.131*** (3.715)	0.122*** (3.763)
Baseline Controls	✓	✓
Firm	✓	✓
State × Year	✓	✓
Observations	33,432	33,055
Adj. R ²	0.639	0.655

This table reports the results of other robustness checks. Panel A reports the results of sub-group analysis based on cash holdings; Panel B reports the results of robustness checks with alternative biodiversity risk measures; Panel C reports the results with alternative trade credit measures; Panel D reports the results with additional controls; Panel E reports the results with alternative fixed effects. **High Cash/Assets** indicates the sub-group that incorporates firms with above-median cash holdings; **Low Cash/Assets** indicates the sub-group that incorporates firms with below-median cash holdings. *Payable/COGS* is accounts payable to cost of goods sold (COGS); *Payable/Purchases* is accounts payable to purchases, where purchases are measured as COGS plus the change of inventory; *Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences; *Biodiversity Mention* is an indicator coded as one if a company mentions biodiversity in at least two sentences; *Biodiversity Regulation* is an indicator equaling one if a company mentions biodiversity in at least two sentences, with at least one referring to regulation; *Nature Dependence* is the measure of revenue-weighted dependence of a firm's economic activities across ecosystem services, developed by Garel et al., (2025); *Adj. Payable/COGS* is accounts payable to cost of goods sold (COGS) adjusted by the industry median; *Adj. Payable/Purchases* is accounts payable to purchases, where purchases are measured as COGS plus the change of inventory, adjusted by the industry median; *Residual Payable/COGS* is the residuals from regressing *Payable/COGS* on climate risk measure (*CC Exposures*) developed by Sautner et al. (2023) by industry and year; *Residual Payable/Purchases* is the residuals from regressing *Payable/Purchases* on climate risk measure (*CC Exposures*) developed by Sautner et al. (2023) by industry and year; *CC Exposures* is the measure of climate risk exposures developed by Sautner et al. (2023); *FA Maturity* is fixed asset maturity measured as the net property, plant, and equipment to depreciation and amortization; *Turnover* is total sales to total assets minus accounts receivable; *Capex/Assets* is capital expenditures to total assets; *R&D/Assets* is research and development expenditures to total assets; *Financing Costs* is total interest expenses to total debts; *Life Cycle* is retained earnings to the book value of total equity; *Financial Constraints* is the KZ index computed according to Kaplan and Zingales (1997); *Bankruptcy Risk* is the Z-score computed according to Altman (1968). *Baseline Controls* is a battery of control variables as defined in Section 3.2.3. See Appendix A for variable definitions. *t*-statistics are in parentheses. *** 1% significance; ** 5% significance; * 10% significance.

costs (Garel et al., 2025). This measure is derived from a combination of segment-level revenue data and Dependency Materiality Ratings from the ENCORE database,²² which aligns closely with other biodiversity measures such as biodiversity footprint (Garel et al., 2024) and reflects a firm's exposure to biodiversity risk. We rely on the revenue-weighted dependence measure, which helps to alleviate bias from selective or delayed disclosure practices. The results presented in Columns (5) and (6) in Panel B, show that our findings remain robust when using this alternative measure despite a slight reduction in the statistical significance, likely due to the smaller sample size and shorter coverage period (2010–2022) of the nature dependence dataset.

We also consider that our results may be influenced by varying trade credit patterns across industries, as firms in certain industries can rely more heavily on trade credit financing than others can. To address this, we adjust the trade credit measures using the industry median value (FF48). Columns (1) and (2) of Panel C demonstrate that our findings remain robust after accounting for the trade credit granted to a firm relative to its industry norms. In addition, firms receiving less trade credit may undermanage byproducts in their production processes due to increased financing pressure. To address potential reverse causality, we lag all independent variables by one year. Columns (3) and (4) of Panel C demonstrate that our results remain robust when using one-period forward trade credit,

²² See <https://www.encorenature.org/en/data-and-methodology/methodology>.

thereby mitigating bias from reverse causality.

Furthermore, Cao et al. (2024) document that climate change uncertainty shapes firms' access to trade credit. Giglio et al. (2023, p.11) highlight that "climate and biodiversity risk are related but distinct concepts". To isolate the unique explanatory power of biodiversity risk beyond climate-related factors, we conduct a two-step regression: first, we regress accounts payable on the climate risk measure developed by Sautner et al. (2023); second, we use the residual from this regression as the dependent variable in our baseline model. Columns (5) and (6) of Panel C show consistent results, suggesting that corporate biodiversity risk exposure captures incremental risk factors beyond those associated with firm-level exposure to climate risk.

Moreover, we incorporate additional control variables identified in prior literature as relevant to trade credit (Cao et al., 2024; Zaman et al., 2025) to mitigate potential bias from omitted factors. Specifically, we control for asset maturity, turnover, capital expenditures, R&D expenditures, financing costs, corporate life cycle, financial constraints, and bankruptcy risk. Additionally, we incorporate climate risk measure in this one-step regression to address the potential bias in our two-step regression using residuals as the dependent variable (Chen et al., 2018).²³ Panel D of Table 6 shows that our findings remain consistent when including additional control variables.

Finally, we employ state-by-year fixed effects to control for time-varying factors specific to the state in which a firm is headquartered. As shown in Panel E of Table 6, We find that our results remain robust when including regional fixed effects.²⁴

4.5. Mechanism tests

In this subsection, we investigate the mechanisms through which higher biodiversity risk leads to reduced access to trade credit. As discussed in the hypothesis development section, we expect that biodiversity risk adversely affects firms' operating performance, thereby driving the reduced access to trade credit. To examine this mechanism, we test whether firms with higher biodiversity risk are more likely to experience decreased operating performance, measured as operating returns and operating expenses, deflated by total assets. We then rerun Equation (1) with these measures as dependent variables.

Table 7 shows that the coefficient of biodiversity risk is significantly negative for operating returns and significantly positive for operating expenses. These results suggest that firms with higher biodiversity risk are more likely to experience reduced operating performance, which can disrupt supply chains and production processes. Therefore, suppliers are likely to take customers' biodiversity risk into account when making trade credit decisions.

4.6. Cross-Sectional analyses

Our results thus far indicate that biodiversity risk adversely impacts firms' access to trade credit. In this subsection, we take an additional step to examine how external competitive forces, specifically product market competition and industry exposure, as well as internal management factors, such as corporate culture and product innovation, influence the observed effect of biodiversity risk on firms' access to trade credit. These tests aim to provide further insights into the cross-sectional differences among firms, highlighting how specific firm characteristics influence the extent to which biodiversity risk constrains access to trade credit.

4.6.1. Product market competition

Previous studies suggest that buyers' market power may influence the trade credit offered by suppliers. For example, Fabbri and Klapper (2016) find that suppliers are more likely to offer trade credit and favorable credit terms to powerful and important customers, who, in turn, are more likely to negotiate longer payment periods and generate overdue payments. Gonçalves et al. (2018) document that customers with greater market power tend to receive longer payment periods as such firms may require additional time from their suppliers to verify the product, while the impact of market power diminishes during credit crunches induced by the global financial crisis. Therefore, we expect suppliers to be more inclined to provide more trade credit to firms highly exposed to biodiversity risks with greater market power.

To examine the effect of market power on the relationship between biodiversity risk and access to trade credit, we use the firm-specific, text-based measures of product market competition developed by Hoberg and Phillips (2016) in the context of 10-Ks, specifically the Herfindahl-Hirschman Index (*HHI*) and the Similarity Index (*SM*). Firms are then divided into high- and low-competition groups based on the median values of these measures. A higher *HHI* (*SM*) indicates a lower (higher) level of product market competition. Table 8 demonstrates that biodiversity risk exposure is associated with less trade credit only when firms operate in highly competitive markets, as shown in Columns (2), (3), (6), and (7). This finding suggests that suppliers are less restrictive in providing trade credit to firms exposed to biodiversity risk when these firms possess greater market power. The coefficients for the biodiversity risk measure in the high-competition groups are approximately four times larger than those in the low-competition groups, although the differences are only statistically significant when accounts payable to COGS is used as the dependent variable.

²³ Additionally, we control for environmental performance proxied by Refinitiv's rating, which significantly reduces our sample size. Our baseline results remain robust (untabulated), indicating that our findings capture the incremental effect of biodiversity risk beyond general environmental performance.

²⁴ To control for regional differences, we included an indicator variable *Different State*, that equals one if the supplier and customer are headquartered in different states in the analysis based on the supplier-customer pair. Our results remain robust. We further add supplier (customer) state-by-year fixed effects to the analysis, and the results are consistent. These results are not tabulated for brevity.

Table 7
Mechanism Tests.

	(1) Operating Returns	(2) Operating Expenses
Biodiversity Negative	−0.009*** (−3.323)	0.035*** (3.315)
Intercept	0.441*** (9.172)	0.632*** (3.191)
Baseline Controls	✓	✓
Firm	✓	✓
Year	✓	✓
Observations	34,431	34,431
Adj. R ²	0.838	0.895

This table reports the results of mechanism tests. *Operating Returns* is the operating income to total assets; *Operating Expenses* is the operation expenses to total assets; *Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences. *Baseline Controls* is a battery of control variables as defined in Section 3.2.3. See Appendix A for variable definitions. *t*-statistics are in parentheses. *** 1% significance; ** 5% significance; * 10% significance.

Table 8
Results of cross-sectional analysis: product market competition.

	High HHI	Low HHI	High SM	Low SM	High HHI	Low HHI	High SM	Low SM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Payable/ COGS	Payable/ COGS	Payable/ COGS	Payable/ COGS	Payable/ Purchase	Payable/ Purchase	Payable/ Purchase	Payable/ Purchase
Biodiversity Negative	−0.015	−0.059***	−0.057***	−0.012	−0.014	−0.041***	−0.040***	−0.011
	(−0.849)	(−2.922)	(−2.838)	(−0.787)	(−0.916)	(−2.853)	(−2.824)	(−0.827)
Intercept	−0.055 (−0.587)	0.243 (1.423)	0.205 (1.125)	−0.111 (−1.112)	−0.052 (−0.574)	0.129 (0.848)	0.129 (0.787)	−0.113 (−1.207)
Baseline Controls	✓	✓	✓	✓	✓	✓	✓	✓
Firm	✓	✓	✓	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓	✓	✓	✓
Wald Test (p-value)	0.066		0.047		0.166		0.106	
Observations	15,188	15,188	15,408	15,408	15,067	14,993	15,203	15,297
Adj. R ²	0.796	0.633	0.607	0.796	0.789	0.651	0.629	0.787

This table reports the results of sub-group analysis based on product market competition. **High HHI** indicates the sub-group that incorporates firms with above-median Herfindahl-Hirschman Index (*HHI*); **Low HHI** indicates the sub-group that incorporates firms with below-median Herfindahl-Hirschman Index (*HHI*); **High SM** indicates the sub-group that incorporates firms with above-median Similarity Index (*SM*); **Low SM** indicates the sub-group that incorporates firms with below-median Similarity Index (*SM*). *Payable/COGS* is accounts payable to cost of goods sold (*COGS*); *Payable/Purchases* is accounts payable to purchases, where purchases are measured as *COGS* plus the change of inventory; *Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences. *Baseline Controls* is a battery of control variables as defined in Section 3.2.3. See Appendix A for variable definitions. *t*-statistics are in parentheses. *** 1% significance; ** 5% significance; * 10% significance.

4.6.2. Industry-Level exposure to biodiversity risks

Firm exposure to physical environmental risks, including biodiversity risks, is influenced by industry norms. While some industries, such as accommodation and telecommunications, might not be directly exposed to biodiversity risk, other industries, such as pharmaceuticals, forestry, and fishery, are highly dependent on biodiversity resources (Giglio et al., 2023; Ma et al., 2024). This might cause the outcomes of biodiversity risk exposure to differ across industries. For example, Bach et al. (2024) document that the effect of biodiversity exposure on overall firm performance differs across industries. Similarly, a recent study by Créti et al. (2024) shows that the effect of biodiversity trends on market pricing will not be reflected in sectors with lower biodiversity risk exposure. Therefore, we expect our results to be more pronounced in industries with higher biodiversity risk exposure, where these issues are more material.

We proxy for industry-level biodiversity risk exposure (*Industry Exposures*) based on the holding-based measure developed by Giglio et al. (2023).²⁵ We divide the sample based on the median value of *Industry Exposures*. Table 9 shows that biodiversity risk exposure is associated with less trade credit only when firms are in highly exposed industries. This suggests that suppliers consider industry norms when deciding whether to grant trade credit to firms exposed to higher biodiversity risks. The Wald test confirms that the difference between the coefficients in the high- and low-exposure industry groups is statistically significant.

²⁵ In addition to the holding-based measure, Giglio et al. (2023) also develop industry-level biodiversity risk measures based on survey data and CDP reports. Our results remain consistent when using these two alternative measures.

Table 9

Results of cross-sectional analysis: industry-level exposure.

	High Industry Exposure	Low Industry Exposure	High Industry Exposure	Low Industry Exposure
	(1)	(2)	(3)	(4)
	Payable/COGS	Payable/COGS	Payable/Purchase	Payable/Purchase
Biodiversity Negative	−0.041***	0.006	−0.029***	0.005
	(−2.802)	(1.255)	(−2.833)	(1.015)
Intercept	0.057	0.256**	−0.048	0.254**
	(0.488)	(2.115)	(−0.453)	(2.150)
Baseline Controls	✓	✓	✓	✓
Firm	✓	✓	✓	✓
Year	✓	✓	✓	✓
Wald Test (p-value)	0.001		0.002	
Observations	21,217	12,834	21,054	12,633
Adj. R ²	0.641	0.668	0.659	0.671

This table reports the results of sub-group analysis based on industry-level exposure. **High Industry Exposure** indicates the sub-group that incorporates firms with above-median holding-based industry-level biodiversity risk exposure measure developed by Giglio et al. (2023); **Low Industry Exposure** indicates the sub-group that incorporates firms with below-median holding-based industry-level biodiversity risk exposure measure developed by Giglio et al. (2023). *Payable/COGS* is accounts payable to cost of goods sold (COGS); *Payable/Purchases* is accounts payable to purchases, where purchases are measured as COGS plus the change of inventory; *Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences. *Baseline Controls* is a battery of control variables as defined in Section 3.2.3. See Appendix A for variable definitions. *t*-statistics are in parentheses. *** 1% significance; ** 5% significance; * 10% significance.

4.6.3. Corporate culture

Suppliers, as an important stakeholder group, may offer favorable trade credit terms based on strong stakeholder relationships. Firms with a more transparent information environment and a long-term orientation may face less severe stakeholder-agency conflicts, thereby securing greater access to trade credit. In this subsection, we examine the moderating effect of corporate culture, as firms with strong cultures are more likely to operate efficiently, prioritize long-term outcomes, and exhibit higher levels of information transparency (K. Li et al., 2021). Furthermore, shared beliefs and values (i.e., stronger corporate culture) enhance communication with stakeholders (Van den Steen, 2010), which solidifies more positive relationships and reduces stakeholder violations (Zaman, 2024). Consequently, firms with a stronger corporate culture may earn greater trust from upstream suppliers, resulting in increased access to trade credit.

We measure corporate culture (*Culture*) using the aggregated firm-specific text-based measure developed by K. Li et al. (2021), based on earnings conference calls. We then divide the sample by the median value of *Culture*. The results presented in Table 10 indicate that biodiversity risk exposure is associated with less access to trade credit among firms with a weak culture, although the difference in the coefficients is significant only when trade credit is measured using accounts payable to COGS. This suggests that suppliers may be more generous in extending trade credit to firms with a strong culture, potentially due to stronger stakeholder relationships fostered by greater corporate transparency and a long-term orientation.

4.6.4. Product innovation

We next examine the role of product innovation in mitigating the relationship between biodiversity risk exposure and access to trade credit. Product innovation is closely tied to capital expenditure on R&D investments (Hoberg & Maksimovic, 2022), which generates new growth opportunities. Furthermore, firms can strategically reduce their exposure to biodiversity transition risks and sustain long-term growth by integrating sustainability practices and ecological considerations into their product development and production processes during the product innovation stage (Bach et al., 2024). As a result, suppliers may consider growth opportunities and the role of product innovation in addressing transition biodiversity risks when determining trade credit terms for highly exposed firms.

We utilize a text-based measure of firm-level product life cycle developed by Hoberg and Maksimovic (2022) in the 10-K context (*Product Innovation*), which captures the degree of a firm being exposed to product innovation stage (stage one) of investment projects, where Tobin's *q* is highly sensitive to capital expenditures and R&D investments. The sample is then divided based on the median value of *Product Innovation*. Table 11 demonstrates that biodiversity risk exposure is associated with less trade credit only in the low product innovation group, with significantly different coefficients between the groups. This suggests that suppliers may perceive high-exposed firms with higher levels of product innovation as more capable of seizing growth opportunities and addressing biodiversity transition risks in production.

4.7. Additional Analysis: Supply chain spillover effects

Our baseline results suggest that suppliers take customers' biodiversity risk exposure into account when offering trade credit. In this section, we examine whether the reduction in trade credit financing spills over into the trade credit supply chain. Specifically, we investigate whether highly exposed firms that receive less trade credit from their suppliers extend less trade credit to their own customers. Based on the specification of Zaman et al. (2025), we examine this potential spillover effect using the following equation:

Table 10

Results of Cross-Sectional Analysis: Corporate Culture.

	Strong Culture	Weak Culture	Strong Culture	Weak Culture
	(1)	(2)	(3)	(4)
	Payable/COGS	Payable/COGS	Payable/Purchase	Payable/Purchase
Biodiversity Negative	0.002	−0.050**	−0.004	−0.030**
	(0.075)	(−2.285)	(−0.169)	(−2.253)
Intercept	−2.187	−0.920	−1.701	−1.242
	(−1.215)	(−0.483)	(−1.203)	(−0.718)
Baseline Controls	✓	✓	✓	✓
Firm	✓	✓	✓	✓
Year	✓	✓	✓	✓
Wald Test (p-value)	0.093		0.324	
Observations	12,017	12,017	11,870	11,906
Adj. R ²	0.635	0.778	0.659	0.778

This table reports the results of sub-group analysis based on corporate culture. **Strong Culture** indicates the sub-group that incorporates firms with above-median corporate culture measure developed by K. Li et al. (2021); **Weak Culture** indicates the sub-group that incorporates firms with below-median corporate culture measure developed by K. Li et al. (2021). *Payable/COGS* is accounts payable to cost of goods sold (COGS); *Payable/Purchases* is accounts payable to purchases, where purchases are measured as COGS plus the change of inventory; *Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences. *Baseline Controls* is a battery of control variables as defined in Section 3.2.3. See Appendix A for variable definitions. *t*-statistics are in parentheses. *** 1% significance; ** 5% significance; * 10% significance.

Table 11

Results of cross-sectional analysis: product innovation.

	High Product Innovation	Low Product Innovation	High Product Innovation	Low Product Innovation
	(1)	(2)	(3)	(4)
	Payable/COGS	Payable/COGS	Payable/Purchase	Payable/Purchase
Biodiversity Negative	0.009	−0.054***	0.008	−0.039***
	(1.359)	(−3.060)	(1.226)	(−3.125)
Intercept	0.199	−0.144	0.170	−0.183
	(1.156)	(−1.206)	(1.146)	(−1.635)
Baseline Controls	✓	✓	✓	✓
Firm	✓	✓	✓	✓
Year	✓	✓	✓	✓
Wald Test (p-value)	0.000		0.000	
Observations	15,205	15,205	15,205	15,205
Adj. R ²	0.567	0.755	0.588	0.756

This table reports the results of sub-group analysis based on product innovation. **High Product Innovation** indicates the sub-group that incorporates firms with above-median product innovation measure developed by Hoberg and Maksimovic (2022); **Low Product Innovation** indicates the sub-group that incorporates firms with below-median product innovation measure developed by Hoberg and Maksimovic (2022). *Payable/COGS* is accounts payable to cost of goods sold (COGS); *Payable/Purchases* is accounts payable to purchases, where purchases are measured as COGS plus the change of inventory; *Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences. *Baseline Controls* is a battery of control variables as defined in Section 3.2.3. See Appendix A for variable definitions. *t*-statistics are in parentheses. *** 1% significance; ** 5% significance; * 10% significance.

$$AR_{i,t} = \alpha + \beta_b Biodiversity_{i,t} + \beta_a AP_{i,t} + \beta_{ba} Biodiversity_{i,t} \times AP_{i,t} + \beta_x X_{i,t} + \delta_i + \nu_t + \varepsilon_{i,t}, \quad (4)$$

where $AR_{i,t}$ is the measures of trade credit granted to customers by firm i at year t , measured as accounts receivable to sales (*Receivable/Sales*) (Cao et al., 2024); $AP_{i,t}$ is the measures of trade credit received from suppliers for firm i at year t (*Payable/COGS*)²⁶; $Biodiversity_{i,t}$ is the measure of negative biodiversity risk exposure (*Biodiversity Negative*) for firm i at year t . $X_{i,t}$ denotes a battery of controls as defined in Section 3.2.3 and Appendix A. We also include firm and year fixed effects, and cluster standard errors at the firm level.

Column (1) of Table 12 show that biodiversity risk exposure is associated with lower levels of trade credit extended to customers. More importantly, the significantly negative coefficient of the interaction term *Biodiversity Negative* $\times$ *Payable/COGS*, as shown in Column (2), suggests that highly exposed firms receiving more conservative trade credit from suppliers extend less generous trade credit to their own customers. This additional finding provides clear evidence that biodiversity risk exposure results in a trade credit supply chain spillover effect. This implies an effect of biodiversity risk on both the upstream and downstream of the supply chain, and, in turn, represents a threat to the stability of the supply chain.

²⁶ For brevity, we only use accounts payable to COGS in this analysis. The results are qualitatively similar if we use accounts payable to purchases.

Table 12
Results of additional analysis: supply chain spillover effects.

	(1) Receivable/Sales	(2) Receivable/Sale
Biodiversity Negative	−0.007**	0.005
	(−2.169)	(1.396)
Payable/COGS		0.109***
		(8.200)
Biodiversity Negative × Payable/COGS		−0.029***
		(−3.766)
Intercept	0.360***	0.344***
	(6.558)	(6.507)
Baseline Controls	✓	✓
Firm	✓	✓
Year	✓	✓
Observations	34,280	34,280
Adj. R ²	0.676	0.687

This table reports the results of additional analysis on supply chain spillover effects. *Receivable/Sales* is accounts receivable to total sales; *Payable/COGS* is accounts payable to cost of goods sold (COGS); *Biodiversity Negative* is the difference between the number of negative biodiversity sentences and positive biodiversity sentences. *Baseline Controls* is a battery of control variables as defined in Section 3.2.3. See Appendix A for variable definitions. *t*-statistics are in parentheses. *** 1% significance; ** 5% significance; * 10% significance.

5. Conclusion

This study examines the relationship between firm-level exposure to biodiversity risk and access to trade credit. Utilizing the text-based measure of firm-level biodiversity risk provided by Giglio et al. (2023), we find that firms' access to trade credit from suppliers decreases significantly with increased firm-level exposure to biodiversity risk. This effect is also economically significant. Moreover, the results hold when addressing the endogeneity issue using an instrumental variable (IV) method, difference-in-differences regression, and analysis based on supplier-customer pairs. Our results are robust when (1) ruling out the alternative explanation of decreased trade credit demand; (2) using alternative biodiversity risk measures; (3) using alternative trade credit measures; and (3) including additional control variables and fixed effects.

The mechanism test shows that reduced operating performance is the primary channel through which biodiversity risk affects access to trade credit. In additional tests, we find that biodiversity risk exposure is associated with less trade credit extended by suppliers mainly in firms (1) with low market power; (2) in highly exposed industries; (3) with weak corporate culture; and (4) with low product innovation. Moreover, we find that firms highly exposed to biodiversity risk that receive less trade credit from suppliers tend to extend less trade credit to their customers, suggesting that biodiversity risk exposure results in a spillover effect within supply chain.

We contribute to two main streams of the literature. First, we contribute to the emerging studies on biodiversity risk, which document several economic consequences of biodiversity risk (e.g., Bach et al., 2024; Cr  ti et al., 2024; Giglio et al., 2023; Y. Li et al., 2025; Liang et al., 2024; Soylemezgil & Uzmanoglu, 2024, among others), by showing that biodiversity risk exposure also affects firms' access to trade credit. This finding is important as trade credit is a crucial source of short-term financing (Hasan & Habib, 2019; Petersen & Rajan, 1997). Second, we contribute to studies on the determinants of trade credit (e.g., El Ghouli & Zheng, 2016; Fabbri & Klapper, 2016; Garc  a-Teruel et al., 2014; Lawrenz & Oberndorfer, 2018; Wu et al., 2014), by documenting another important factor influencing firms access to trade credit, biodiversity risk exposure, which represents an increasingly significant risk factor that could impose both physical and transitional concerns on firms.

Our findings provide several practical implications. First, our findings highlight that firm-level exposure to biodiversity risk affects access to trade credit and that industry exposure further affects such access. This finding has implications for policymakers in terms of considering the impact of biodiversity risk at the firm level when formulating policies, as transition risk poses a significant challenge for firms and directly affects their access to trade credit. This is particularly important, as this source of financing helps firms manage liquidity risk. Moreover, policymakers may need to adequately address the complexities introduced by biodiversity risk and the role that industry norms play in such risk.

Further, our results highlight that suppliers consider biodiversity-related risk when extending trade credit, which indicates increased awareness of biodiversity risk, not only by investors but also by different stakeholders. Moreover, this finding reveals the complex interplay between environmental considerations and corporate liquidity. Thus, our study provides insights to managers, as it underscores the importance of managing biodiversity risk exposure, particularly for financially constrained firms and those in industries with high biodiversity risk exposure, as such risk further hinders their ability to access trade credit. Given the findings of prior studies that the management of biodiversity risk could mitigate its negative consequences (Bassen et al., 2024), it has become increasingly important for firms to integrate biodiversity into corporate strategies and actions.

Our study opens several avenues for future research. Due to data limitations, our study primarily examines the impact of biodiversity risk on firms' access to trade credit, which is a crucial source of financing, and whether this effect extends to the firm's own customers. Future research could explore how biodiversity risk affects the upstream supply chain, particularly whether firms respond

to biodiversity-related disruptions by terminating relationships with existing suppliers and transitioning to alternative sources. Additionally, while we account for the geographic factors that could affect the association between biodiversity risk and access to trade credit, future research could examine additional dimensions such as physical distance, culture, language, and beliefs within supplier-customer relationships in an international setting.

CRedit authorship contribution statement

Khadija S. Almaghrabi: Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization. **Walid Ben-Amar:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Ziyu Kong:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A.: Variable definition

Variables	Definition	Source
Dependent Variables		
Payable/COGS	Accounts payable (ap) to cost of goods sold (cogs).	Compustat
Payable/Purchases	Accounts payable (ap) to purchases (cogs + d.invt).	Compustat
Adj. Payable/COGS	Accounts payable (ap) to cost of goods sold (cogs) subtracting the industry median. The industry classification follows the Fama-French 48 industry classification scheme.	Compustat
Adj. Payable/Purchases	Accounts payable (ap) to purchases (cogs + d.invt) subtracting the industry median. The industry classification follows the Fama-French 48 industry classification scheme.	Compustat
Residual Payable/COGS	The residuals from regressing accounts payable (ap) to cost of goods sold (cogs) on climate risk by industry and year, where climate risk is measured using a firm-specific, text-based metric developed by Sautner et al. (2023) .	Compustat; Sautner et al. (2023) [2001 – 2023]
Residual Payable/Purchases	The residuals from regressing accounts payable (ap) to purchases (cogs + d.invt) on climate risk by industry and year, where climate risk is measured using a firm-specific, text-based metric developed by Sautner et al. (2023) .	Compustat; Sautner et al. (2023) [2001 – 2023]
Operating Returns	Operating incomes (oibdp) to total assets (at).	Compustat
Operating Expenses	Operating expenses (xopr) to total assets (at).	Compustat
Receivable/Sales	Accounts receivable (rect) to sales (sale).	Compustat
Independent Variables		
Biodiversity Negative	The difference between the number of negative biodiversity sentences and positive biodiversity sentences.	Giglio et al. (2023)
Biodiversity Mention	A binary variable which equals one if a company mentions biodiversity in at least two sentences in its 10-K.	Giglio et al. (2023)
Biodiversity Regulation	A binary variable which equals one if a company mentions biodiversity in at least two sentences, with at least one referring to regulation.	Giglio et al. (2023)
Nature Dependence Treat	The revenue-weighted dependence of a firm's economic activities across ecosystem services. An indicator coding one if a firm locates in non-suing states towards the rollback of the Endangered Species Act (ESA) by the Trump administration in 2019. Suing states are Colorado, Connecticut, Illinois, Michigan, Nevada, New Jersey, New Mexico, New York, North Carolina, Oregon, Pennsylvania, Rhode Island, Vermont, Washington, the District of Columbia, Maryland, Massachusetts, and California.	Garel et al. (2025) [2010–2022] Self-constructed
Post	An indicator coding one for the years 2019–2023, zero for the years 2015–2018	Self-constructed
Instrumental Variables		
Google Attention	The aggregate attention indexes that analyze the annual Google searches for the terms “biodiversity loss”, “ecosystem services”, and “species loss”, scaled by 1000.	Google Trends
Moderators		
HHI	The text-based firm-specific Herfindahl–Hirschman index.	Hoberg and Phillips (2016) [2000 – 2021]
SM	The text-based firm-specific Similarity index.	Hoberg and Phillips (2016) [2000 – 2021]
Industry Exposures	The holding-based industry-level biodiversity risk exposures.	Giglio et al. (2023)
Culture	The aggregated text-based corporate culture measure.	K. Li et al. (2021) [2002 – 2021]
Product Innovation	The text-based measure of the degree of a firm being exposed to the first stage of product life cycle.	Hoberg and Maksimovic (2022) [2000 – 2021]

(continued on next page)

(continued)

Variables	Definition	Source
Controls		
Ln(Assets)	The natural logarithm of total assets (at).	Compustat
Ln(Age)	The natural logarithm of firm age, measured as the years since a firm appears in Annual Compustat North American file.	Compustat
PPE/Assets	Net property, plant, and equipment (ppent) to total assets (at).	Compustat
ΔSales	The annual change of sales (sale) divided by lagged sales.	Compustat
ROA	Net income (ni) to total assets (at).	Compustat
MB	The market value of equity (csho*prcc.f) to the book value of equity (seq).	Compustat
Debts/Assets	Total debts (dltt + dlc) to total assets (at).	Compustat
Cash/Assets	Cash and short-term investments (che) to total assets (at).	Compustat
σCash Flow	The standard deviation of cash flows (oibdp – xint – txt – dvp – dvc) to total assets (at) over at least 3 of the last 5 years.	Compustat
Customer Share	A supplier's sales to a specific customer to the supplier's total sales	Compustat
CC Exposure	Firm-specific text-based measure of climate risk exposure.	Sautner et al. (2023) [2001 – 2023]
FA Maturity	Net property, plant, and equipment (ppent) to depreciation and amortization (dp).	Compustat
Turnover	Total sales (sale) to total assets (at) minus accounts receivable (rect).	Compustat
Capx/Assets	Capital expenditures (capx) to total assets (at).	Compustat
R&D/Assets	Research and development expenditures (xrd) to total assets (at).	Compustat
Financing Costs	Total interest expenses (xint) to total debts (dltt + dlc).	Compustat
Life Cycle	Retained earnings (re) to the book value of total equity (seq).	Compustat
Financial Constraints	KZ index computed according to Kaplan and Zingales (1997).	Compustat
Bankruptcy Risk	Z-score computed according to Altman (1968).	Compustat

Data availability

The authors do not have permission to share data.

References

- Abdulla, Y., Dang, V.A., Khurshed, A., 2020. Suppliers' listing status and trade credit provision. *Finance* 60, 101535.
- Acharya, V., Davydenko, S.A., Strebulava, I.A., 2012. Cash holdings and credit risk. *Rev. Financ. Stud.* 25 (12), 3572–3609.
- Adamolekun, G., 2024. Firm biodiversity risk, climate vulnerabilities, and bankruptcy risk. *J. Int. Financ. Mark. Inst. Money* 97, 102075.
- Agostino, M., Trivieri, F., 2019. Does trade credit affect technical efficiency? Empirical evidence from Italian manufacturing SMEs. *J. Small Bus. Manag.* 57 (2), 576–592.
- Altman, E.I., 1968. Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *J. Financ.* 23 (4), 589–609.
- Arnold, T., Tran(Nancy), N., 2024. Biodiversity risk and bond credit ratings. *SSRN Electron. J.* 4947029.
- Bach, T.N., Hoang, K., Le, T., 2024. Biodiversity risk and firm performance. *SSRN Electron. J.* 4735012.
- Barrot, J.-N., Sauvagnat, J., 2016. Input specificity and the propagation of idiosyncratic shocks in production networks. *Q. J. Econ.* 131 (3), 1543–1592.
- Bassen, A., Buchholz, D., Lopatta, K., Rudolf, A.R., 2024. Biodiversity management and stock price crash risk. *Bus. Strateg. Environ.* 33 (5), 4788–4805.
- Beck, T., Demirgüç-Kunt, A., Maksimovic, V., 2008. Financing patterns around the world: are small firms different? *J. Financ. Econ.* 89 (3), 467–487.
- Burkart, M., Ellingsen, T., 2004. In-kind finance: a theory of trade credit. *Am. Econ. Rev.* 94 (3), 569–590.
- Campello, M., Gao, J., 2017. Customer concentration and loan contract terms. *J. Financ. Econ.* 123 (1), 108–136.
- Cao, Z., Chen, S.X., Dong, T., Lee, E., 2024. Climate change uncertainty and supply chain financing. *Br. Account. Rev.* 101423.
- Chen, W., Hribar, P., Melessa, S., 2018. Incorrect inferences when using residuals as dependent variables. *J. Account. Res.* 56 (3), 751–796. <https://onlinelibrary.wiley.com/doi/10.1111/1475-679X.12195>.
- Chong, B.U., Im, H.J., 2024. The effect of affiliation with a large business group on trade credit finance: an empirical analysis of chaebol firms in Korea. *J. Bus. Financ. Acc.* 51 (7–8), 1828–1857.
- Clarkson, M., 1994. A Risk Based Model of Stakeholder Theory. In: *Proceedings of the 2nd Toronto Conference on Stakeholder Theory, Centre for Corporate Social Performance and Ethics, University of Toronto, Toronto, April*.
- Clarkson, M., 1995. A stakeholder framework for analyzing and evaluating corporate social performance. *Acad. Manag. Rev.* 20 (1), 92–117.
- Cosma, S., Rimo, G., Cosma, S., 2023. Conservation finance: what are we not doing? a review and research agenda. *J. Environ. Manage.* 336, 117649.
- Créti, A., Huang, Y., Jiang, B., Sanin, M.E., 2024. Biodiversity risk, firm performance, and market mispricing. *SSRN Electron. J.* 4765039.
- Dasgupta, P., 2021. *The economics of biodiversity: The dasgupta review*. https://assets.publishing.service.gov.uk/media/602e92b2e90e07660f807b47/The_Economics_of_Biodiversity-The_Dasgupta_Review_Full_Report.pdf.
- de Carvalho, S.H.C., Cojoianu, T., Ascui, F., 2023. From impacts to dependencies: a first global assessment of corporate biodiversity risk exposure and responses. *Bus. Strateg. Environ.* 32 (5), 2600–2614.
- Dempsey, J., 2013. Biodiversity loss as material risk: Tracking the changing meanings and materialities of biodiversity conservation. *Geoforum* 45, 41–51.
- Deutz, A., Heal, G. M., Niu, R., Swanson, E., Townshend, T., Li, Z., Delmar, A., Meghji, A., Sethi, S. A., & la Puente, J. T., 2020. Financing nature: Closing the global biodiversity financing gap. In *The Paulson Institute, The Nature Conservancy, and the Cornell Atkinson Center for Sustainability*. https://www.paulsoninstitute.org/wp-content/uploads/2020/10/FINANCING-NATURE_Full-Report_Final-with-endorsements_101420.pdf.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K., 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding (arXiv:1810.04805). *ArXiv*.
- Ding, G., Lei, J., Liu, Y., Wang, Z., 2024. Supplier–customer cultural similarity and supplier performance. *J. Bank. Financ.* 163, 107188.
- Donaldson, T., Preston, L.E., 1995. The stakeholder theory of the corporation: Concepts, evidence, and implications. *Acad. Manag. Rev.* 20 (1), 65.
- D'Mello, R., Toscano, F., 2020. Economic policy uncertainty and short-term financing: the case of trade credit. *Finance* 64, 101686.
- El Ghoul, S., Zheng, X., 2016. Trade credit provision and national culture. *Finance* 41, 475–501.
- Ersahin, N., Giannetti, M., Huang, R., 2024. Trade credit and the stability of supply chains. *J. Financ. Econ.* 155, 103830.
- Fabbri, D., Klapper, L.F., 2016. Bargaining power and trade credit. *Finance* 41, 66–80.

- Fabbri, D., Menichini, A.M.C., 2010. Trade credit, collateral liquidation, and borrowing constraints. *J. Financ. Econ.* 96 (3), 413–432.
- Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. *J. Financ. Econ.* 33 (1), 3–56.
- Fava, S.D., Gupta, R., Pierdzioch, C., Rognone, L., 2024. Forecasting international financial stress: the role of climate risks. *J. Int. Financ. Mark. Inst. Money* 92, 101975.
- Flammer, C., Giroux, T., Heal, G., 2024. Biodiversity Finance. *National Bureau of Econ. Res.* No. 31022.
- Freeman, R.E., 1984. *Strategic management: A stakeholder approach*. Pitman.
- Freeman, R.E., Reed, D.L., 1983. Stockholders and stakeholders: a new perspective on corporate governance. *Calif. Manage. Rev.* 25 (3), 88–106.
- García-Teruel, P.J., Martínez-Solano, P., Sánchez-Ballesta, J.P., 2014. Supplier financing and earnings quality. *J. Bus. Financ. Acc.* 41 (9–10), 1193–1211.
- Garel, A., Romec, A., Sautner, Z., Wagner, A.F., 2024. Do investors care about biodiversity? *Eur. Financ. Rev.* 28 (4), 1151–1186.
- Garel, A., Romec, A., Sautner, Z., & Wagner, A. F., 2025. Firm-level nature dependence. European Corporate Governance Institute – Finance Working Paper No. 1050/2025, Swiss Finance Institute Research Paper No. 25-44, HKU Jockey Club Enterprise Sustainability Global Research Institute Paper No. 2025/036.
- Kunming-Montreal Global Biodiversity Framework (GBF), 2022. Conference of the Parties to the Convention on Biological Diversity. <https://www.cbd.int/doc/decisions/cop-15/cop-15-dec-04-en.pdf>.
- Gertler, M., 1988. Financial structure and aggregate economic activity: An overview. https://www.nber.org/system/files/working_papers/w2559/w2559.pdf.
- Giannetti, M., Burkart, M., Ellingsen, T., 2011. What you sell is what you lend? explaining trade credit contracts. *Rev. Financ. Stud.* 24 (4), 1261–1298.
- Giglio, S., Kuchler, T., Stroebel, J., Zeng, X., 2023. Biodiversity risk. *National Bureau of Econ. Res.* No. 31137.
- Giglio, S., Kuchler, T., Stroebel, J., Wang, O., 2024. The economics of biodiversity loss. *National Bureau of Econ. Res.* No. 32678, 1–64.
- Gonçalves, A.B., Schiozer, R.F., Sheng, H.H., 2018. Trade credit and product market power during a financial crisis. *Finance* 49, 308–323.
- Guer, Y., Mueller, L., Schiereck, D., 2024. Biodiversity: a bibliometric analysis of accounting, economics, and finance journals. *Journal of Alternative Finance* 1 (2), 194–210.
- Hasan, M.M., Habib, A., 2019. Social capital and trade credit. *Int. Rev. Financ. Anal.* 61, 158–174.
- Hertzel, M., Li, Z., Officer, M., Rodgers, K., 2008. Inter-firm linkages and the wealth effects of financial distress along the supply chain. *J. Financ. Econ.* 87 (2), 374–387.
- Hoberg, G., Maksimovic, V., 2022. Product life cycles in corporate finance. *Rev. Financ. Stud.* 35 (9), 4249–4299.
- Hoberg, G., Phillips, G.M., 2016. Text-based network industries and endogenous product differentiation. *J. Polit. Econ.* 124 (5), 1423–1465.
- Houdet, J., Trommetter, M., Weber, J., 2012. Understanding changes in business strategies regarding biodiversity and ecosystem services. *Ecol. Econ.* 73, 37–46.
- Houston, J.F., Lin, C., Zhu, Z., 2016. The financial implications of supply chain changes. *Manag. Sci.* 62 (9), 2520–2542.
- Hutchinson, M.C., Lucey, B., 2024. A bibliometric and systemic literature review of biodiversity finance. *Financ. Res. Lett.* 64 (March), 105377.
- Jacobson, T., von Schedvin, E., 2015. Trade credit and the propagation of corporate failure: an empirical analysis. *Econometrica* 83 (4), 1315–1371.
- Jory, S.R., Khieu, H.D., Ngo, T.N., Phan, H.V., 2020. The influence of economic policy uncertainty on corporate trade credit and firm value. *Finance* 64, 101671.
- Kalhor, M. R., & Kyaw, K., 2024. Manage biodiversity risk exposure? *Finance Research Letters*, 61(August 2023), 104989.
- Kaplan, S., Zingales, L., 1997. Do financing constraints explain why investment is correlated with cash flow? *Q. J. Econ.* 112 (1), 169–216.
- Karolyi, G.A., Tobin-de la Puente, J., 2023. Biodiversity finance: a call for research into financing nature. *Financ. Manag.* 52 (2), 231–251.
- Kedward, K., Ryan-Collins, J., Chenet, H., 2023. Biodiversity loss and climate change interactions: Financial stability implications for central banks and financial supervisors. *Clim. Pol.* 23 (6), 763–781.
- Klapper, L., Laeven, L., Rajan, R., 2012. Trade credit contracts. *Rev. Financ. Stud.* 25 (3), 838–867.
- Kunming Declaration, 2021. Ecological Civilization: Building a Shared Future for All Life on Earth. <https://www.cbd.int/doc/c/c2db/972a/fb32e0a277b1ccff742be5/cop-15-05-add1-en.pdf>.
- Lambert, J., 2019. Trump administration weakens Endangered Species Act. *Nature*.
- Lawrenz, J., Oberndorfer, J., 2018. Firm size effects in trade credit supply and demand. *J. Bank. Financ.* 93, 1–20.
- Levine, R., Lin, C., Xie, W., 2018. Corporate resilience to banking crises: the roles of trust and trade credit. *J. Financ. Quant. Anal.* 53 (4), 1441–1477.
- Lian, Y., 2017. Financial distress and customer-supplier relationships. *Finance* 43, 397–406.
- Liang, C., Yang, J., Shen, L., Dong, D., 2024. The role of biodiversity risk in stock price crashes. *Financ. Res. Lett.* 67 (PA), 105856.
- Li, K., Mai, F., Shen, R., Yan, X., 2021a. Measuring corporate culture using machine learning. *Rev. Financ. Stud.* 34 (7), 3265–3315.
- Li, X., Ng, J., Saffar, W., 2021b. Financial reporting and trade credit: evidence from mandatory IFRS adoption. *Contemp. Account. Res.* 38 (1), 96–128.
- Li, T., Ang, T. C. 'Chewie,' & Lu, C., 2023. Employment protection and the provision of trade credit. *Journal of Banking & Finance*, 155, 106991.
- Li, Y., Liu, X., Canil, J., & Cheong, C. S., 2025. Biodiversity risk and firm efficiency. *Finance Research Letters*, 71(October 2024).
- Liu, Y., Dong, H., Wang, Y., 2025. In the same boat: climate risk and hidden debt in the supply chain. *J. Int. Money Financ.* 153, 103299.
- Ma, F., Wu, H., Zeng, Q., 2024. Biodiversity and stock returns. *Int. Rev. Financ. Anal.* 95 (PA), 103386.
- Nedopil, C., 2023. Integrating biodiversity into financial decision-making: challenges and four principles. *Bus. Strateg. Environ.* 32 (4), 1619–1633.
- Newman, D.J., Cragg, G.M., 2020. Natural products as sources of new drugs over the nearly four decades from 01/1981 to 09/2019. *Journal of Natural Products* 83 (3), 770–803.
- Opler, T., Pinkowitz, L., Stulz, R., Williamson, R., 1999. The determinants and implications of corporate cash holdings. *J. Financ. Econ.* 52 (1), 3–46.
- Panwar, R., 2023. Business and biodiversity: achieving the 2050 vision for biodiversity conservation through transformative business practices. *Biodivers. Conserv.* 32 (11), 3607–3613.
- Petersen, M.A., Rajan, R.G., 1997. Trade credit: Theories and evidence. *Rev. Financ. Stud.* 10 (3), 661–691.
- Quan, X., Xiang, C., Tina Gao, R., 2024. Reputation is golden: Superstar CEOs and trade credit. *Journal of Business Finance & Accounting*, 51(1–2), 631–656.
- Raddatz, C., 2010. Credit chains and sectoral comovement: does the use of trade credit amplify sectoral shocks? *Rev. Econ. Stat.* 92 (4), 985–1003.
- Sautner, Z., Van Lent, L., Vilkov, G., Zhang, R., 2023. Firm-level climate change exposure. *J. Financ.* 78 (3), 1449–1498.
- Soylomezgil, S., Uzmanoglu, C., 2024. Biodiversity risk in the corporate bond market. *SSRN Electron. J.* 4721219.
- Staiger, D., Stock, J.H., 1997. Instrumental variables regression with weak instruments. *Econometrica* 65 (3), 557.
- Stock, J. H., Yogo, M., 2005. Testing for weak instruments in linear IV regression. In Andrews DWK Identification and Inference for Econometric Models. (pp. 80–108). New York: Cambridge University Press.
- Svartzman, R., Espagne, E., Julien, G., Paul, H.-L., Mathilde, S., Allen, T., Berger, J., Calas, J., Godin, A., & Vallier, A., 2021. A “silent spring” for the financial system? Exploring biodiversity-related financial risks in France. *Banque de France Working Paper No. 826*.
- Textile Exchange, 2021. *Biodiversity Insights Report*. <https://textileexchange.org/app/uploads/2021/11/Biodiversity-Insights-Report-2021.pdf>.
- TNFD. (2023). *Recommendations of the Taskforce on Nature-related Financial Disclosures*. <https://tnfd.global/wp-content/uploads/2023/08/Recommendations-of-the-Taskforce-on-Nature-related-Financial-Disclosures.pdf?v=1734112245>.
- Van den Steen, E., 2010. Culture clash: the costs and benefits of homogeneity. *Manag. Sci.* 56 (10), 1718–1738.
- Wang, Y., Wu, Z., Zhang, G., 2022. Firms and climate change: a review of carbon risk in corporate finance. *Carbon Neutrality* 1 (1), 6. <https://link.springer.com/10.1007/s43979-022-00005-9>.
- WEF. (2024). *Global Risks Report*. <https://www.weforum.org/publications/global-risks-report-2024/>.
- World Bank. (2021). <https://www.worldbank.org/en/news/press-release/2021/07/01/protecting-nature-could-avert-global-economic-losses-of-usd2-7-trillion-per-year>.
- Wu, W., Firth, M., Rui, O.M., 2014. Trust and the provision of trade credit. *J. Bank. Financ.* 39 (1), 146–159.
- WWF, 2024. *Living Planet Report 2024 – A System in Peril*. https://www.paulsoninstitute.org/wp-content/uploads/2020/10/FINANCING-NATURE_Full-Report_Final-with-endorsements.101420.pdf.
- WWF. (2022). *Living planet report 2022: Building a nature - positive society*. https://wwfint.awsassets.panda.org/downloads/embargo_13_10_2022_lpr_2022_full_report_single_page_1.pdf.

- Yang, S.A., Birge, J.R., 2018. Trade credit, risk sharing, and inventory financing portfolios. *Manag. Sci.* 64 (8), 3667–3689.
- Zaman, R., 2024. When corporate culture matters: the case of stakeholder violations. *Br. Account. Rev.* 56 (1), 101188.
- Zaman, R., Atawnah, N., Banigidadmath, D., Nadeem, M., Liu, J., 2025. Do companies' green credentials enhance trade credit provisions? Global evidence. *J. Int. Financ. Mark. Inst. Money* 103, 102204.
- Zhang, Y., García Lara, J.M., Tribó, J.A., 2020. Unpacking the black box of trade credit to socially responsible customers. *J. Bank. Financ.* 119, 105908.